CHAPTER 07

Aldehydes and Ketones II: Enolates

Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: 208ea74db3f7f7deaf994126006b6f412727957ab157f8aeed86d166aca93996

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

Coverage checklist

ch07:exercise:alpha-map | alpha-map:stage-1, alpha-map:stage-2, alpha-map:stage-3

ch07:exercise:enolate-control | enolate-control:stage-1, enolate-control:stage-2, enolate-control:stage-3

ch07:exercise:aldol-map | aldol-map:stage-1, aldol-map:stage-2, aldol-map:stage-3

ch07:exercise:michael-site | michael-site:stage-1, michael-site:stage-2, michael-site:stage-3

ch07:lesson:alpha | alpha:rule, alpha:cue, alpha:trap, alpha:app, alpha:step-1, alpha:step-2

ch07:lesson:tautomers | tautomers:rule, tautomers:cue, tautomers:trap, tautomers:app, tautomers:step-1, tautomers:step-2, tautomers:step-3

ch07:lesson:aldol | aldol:rule, aldol:cue, aldol:trap, aldol:app, aldol:step-1, aldol:step-2, aldol:step-3

ch07:lesson:conjugate | conjugate:rule, conjugate:cue, conjugate:trap, conjugate:app, conjugate:step-1, conjugate:step-2

ch07:molecule:ketone | ketone

ch07:molecule:ethanal-enolate | ethanal-enolate

ch07:molecule:vinylalcohol | vinylalcohol

ch07:molecule:aldol | aldol

ch07:molecule:crotonaldehyde | crotonaldehyde

ch07:molecule:methylvinylketone | methylvinylketone

ch07:curriculum | scope-1, scope-2, scope-3

Curriculum scope

scope-1: 7.1 General Principles

scope-2: 7.2 Enolate Chemistry

scope-3: 7.3 Aldol Condensation

Objective: Identify alpha and beta positions

Objective: Distinguish tautomers from resonance and kinetic from thermodynamic control

Objective: Trace aldol and conjugate additions

Supplied Kaplan curriculum: Chapter 7; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter07b.xhtml

Enolate acidity cross-check: OpenStax 22.5 | https://openstax.org/books/organic-chemistry/pages/22-5-acidity-of-alpha-hydrogen-atoms-enolate-ion-formation

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

alpha | Alpha carbons and acidic hydrogens

The conjugate base is resonance stabilized

alpha:rule | Core Rules

An alpha carbon is adjacent to a carbonyl carbon. Removing an alpha H yields an enolate, whose negative charge can be represented on oxygen or alpha carbon. Ordinary aldehyde/ketone alpha H pKa values are roughly 17–21; a 1,3-dicarbonyl can be substantially more acidic.

alpha:cue | Memory Cues

Alpha = one bond from C=O. Carbonyl carbon is not alpha.

alpha:trap | Common Mistakes / MCAT Traps

Only a site bearing H can be deprotonated there. Stabilization of an enolate explains alpha acidity; do not treat every C–H bond in a carbonyl molecule as equally acidic.

alpha:app | MCAT Applications

Enolate-like intermediates allow enzymes to make or break carbon–carbon bonds adjacent to carbonyls.

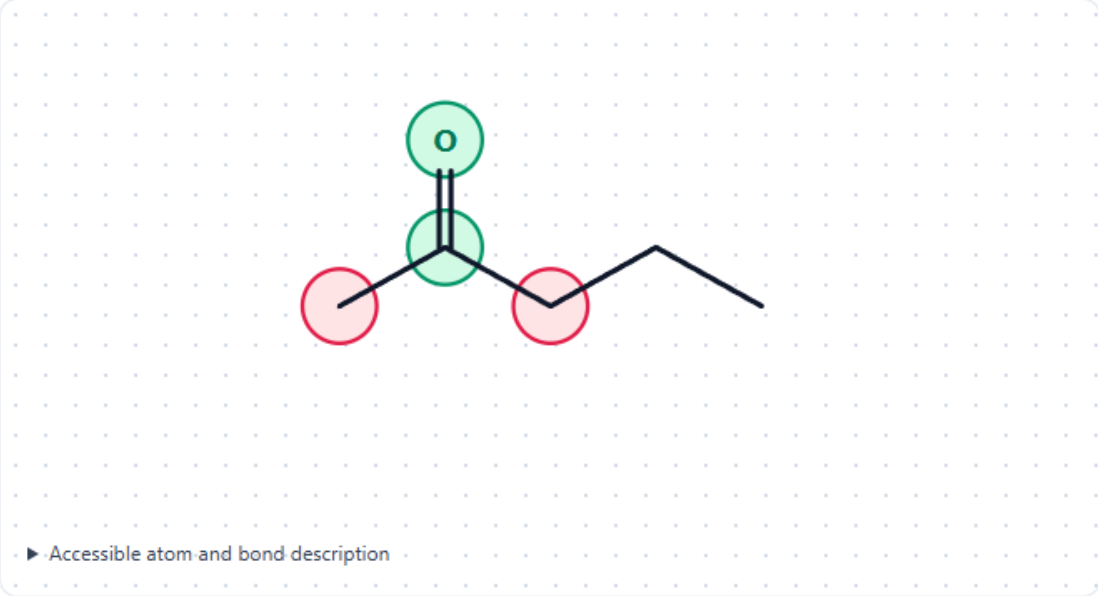
alpha:step-1 | 1. Locate alpha sites

Pentan-2-one has alpha carbons on both sides of its carbonyl: the terminal methyl and the neighboring methylene.

Displayed molecule ID: ketone

{"highlights": {"prin": [1, 5], "alpha": [0, 2]}, "revealName": true}

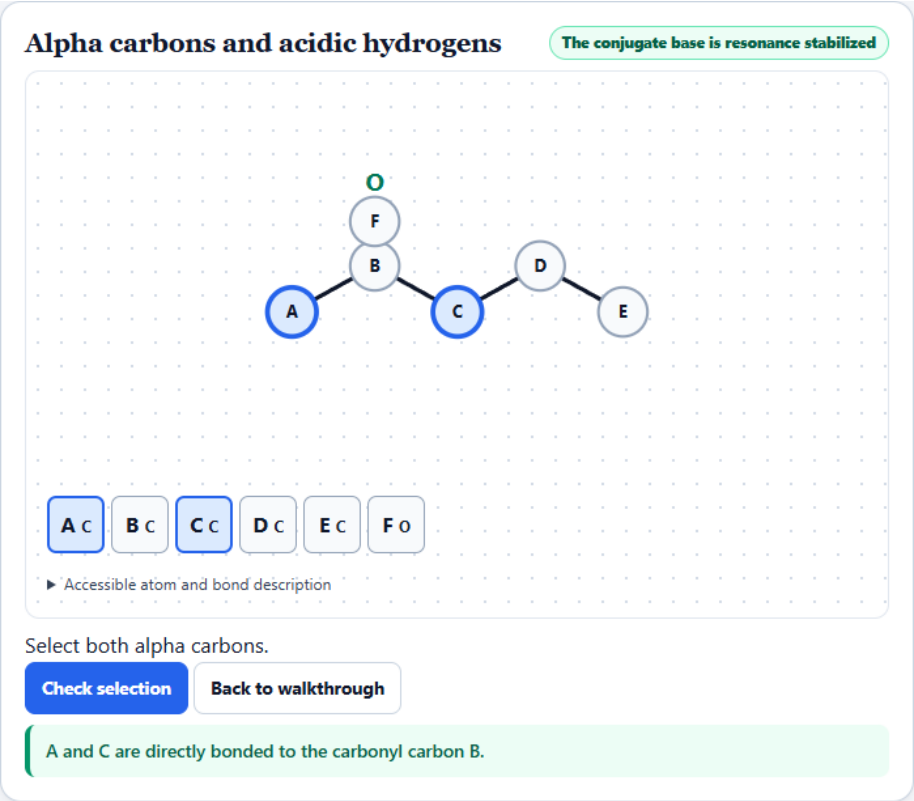
Screenshot: alpha-step-1



Embedded challenge: Select both alpha carbons.

{"prompt": "Select both alpha carbons.", "answer": [0, 2], "explanation": "A and C are directly bonded to the carbonyl carbon B."}

Screenshot: alpha-step-1-challenge



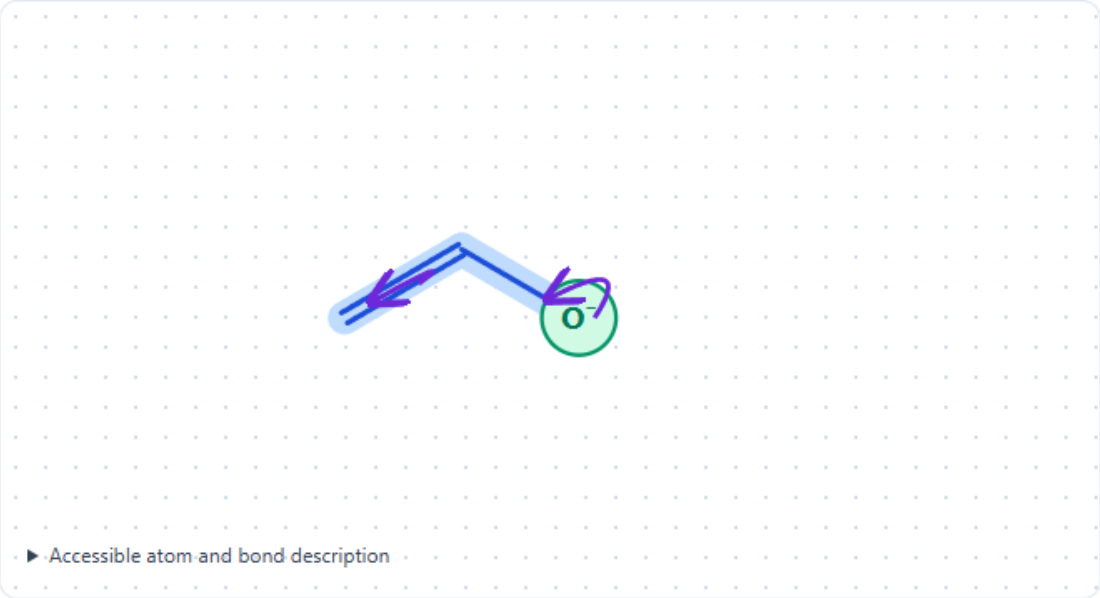
alpha:step-2 | 2. Delocalize charge

The ethanal enolate is drawn with negative charge on O. A second contributor places the charge at the alpha carbon; both describe the same delocalized ion.

Displayed molecule ID: ethanal-enolate

{"highlights": {"path": [0, 1, 2], "prin": [2], "flow": [{"fromAtom": 2, "toBond": 1}, {"fromBond": 0, "toAtom": 0}]}, "revealName": true}

Screenshot: alpha-step-2



tautomers | Tautomers and enolate selectivity

Kinetic control vs. thermodynamic control

tautomers:rule | Core Rules

Keto and enol forms are constitutional isomers that differ in H placement and a double bond; interconversion requires proton transfer. An enolate is the deprotonated species, not another neutral tautomer. Bulky strong base at low temperature under irreversible conditions often favors the less substituted kinetic enolate. Reversible conditions often favor the more substituted thermodynamic enolate.

tautomers:cue | Memory Cues

Tautomers move H and pi bonds. Resonance moves only electrons.

tautomers:trap | Common Mistakes / MCAT Traps

Kinetic/thermodynamic predictions require the substrate and conditions. More substituted is not a universal stability rule for every enolate. Simple aldehydes and ketones usually favor the keto form, but conjugation and aromaticity can change that.

tautomers:app | MCAT Applications

Selective enolate generation lets a synthesis control which alpha carbon forms a new bond.

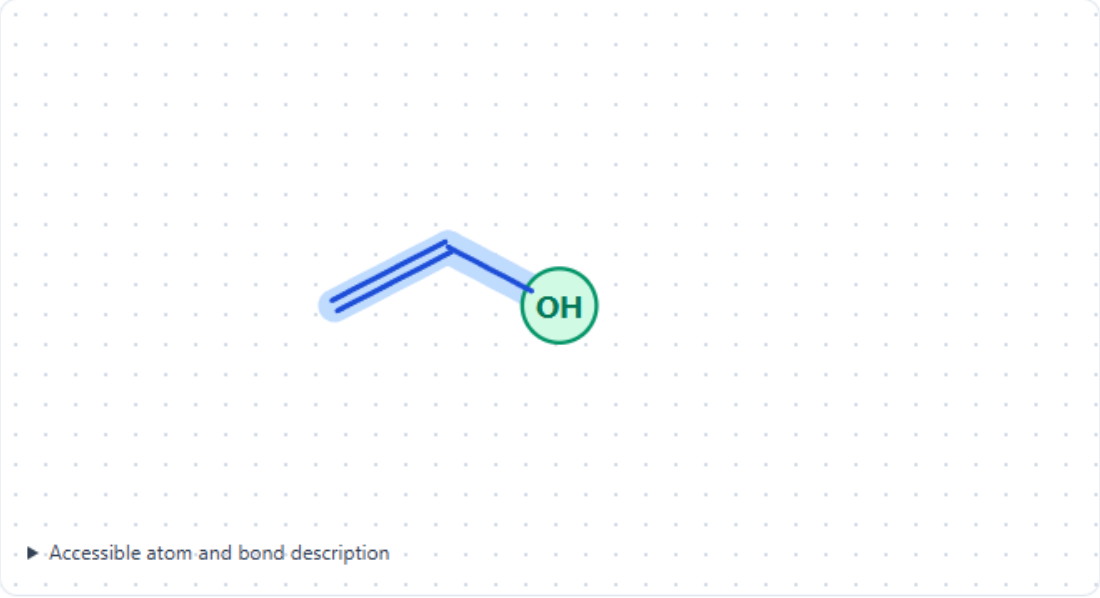
tautomers:step-1 | 1. Keto vs. enol

Ethanal and ethenol have the same formula but different H and bond placement. This enol has OH directly attached to an alkene carbon.

Displayed molecule ID: vinylalcohol

{"highlights": {"path": [0, 1, 2], "prin": [2]}, "revealName": true}

Screenshot: tautomers-step-1



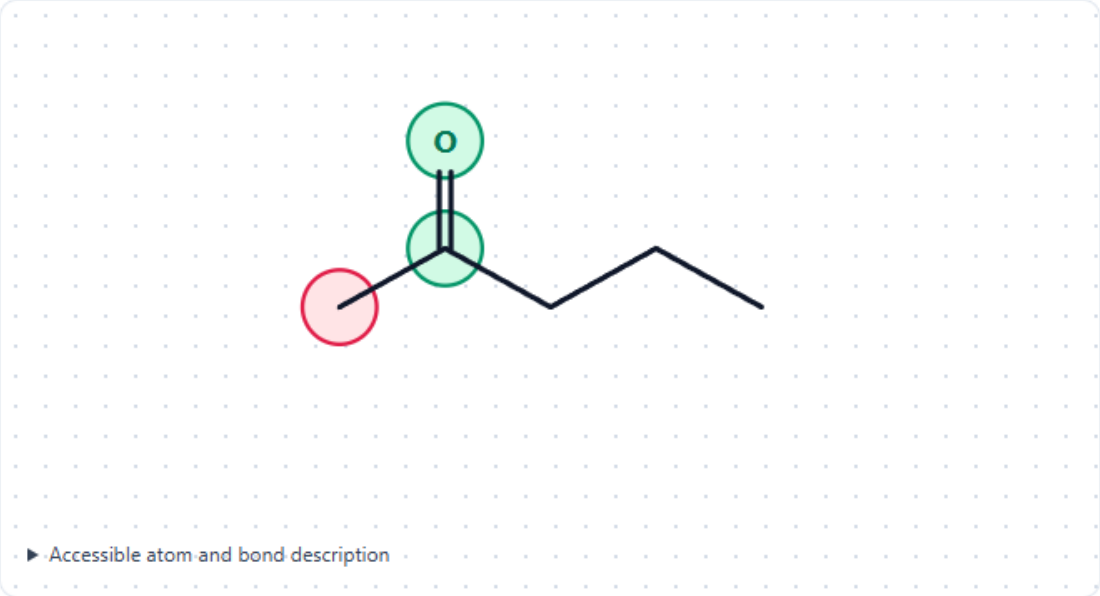
tautomers:step-2 | 2. Kinetic site

For pentan-2-one, bulky LDA under typical low-temperature kinetic conditions often removes an H from the less hindered methyl side.

Displayed molecule ID: ketone

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Screenshot: tautomers-step-2



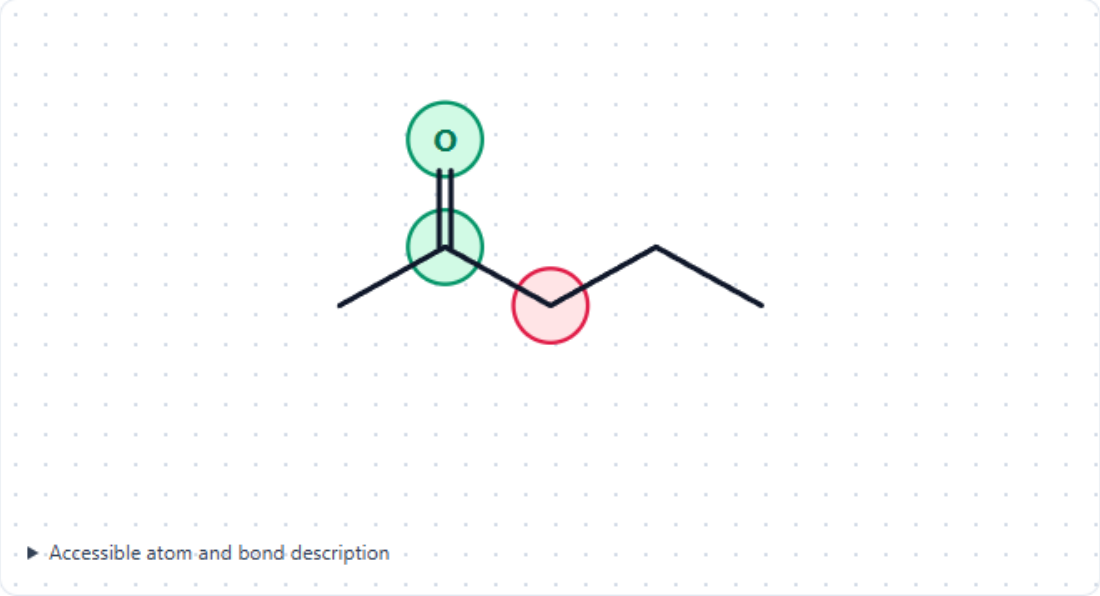
tautomers:step-3 | 3. Thermodynamic site

Under suitable reversible conditions, the more substituted enolate from the methylene side is often favored. State the conditions before predicting selectivity.

Displayed molecule ID: ketone

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Screenshot: tautomers-step-3



aldol | Aldol addition, condensation, and retro-aldol

New bond: donor alpha C to acceptor carbonyl C

aldol:rule | Core Rules

An enolate adds to another carbonyl and is protonated to give a beta-hydroxy aldehyde or ketone: aldol addition. Subsequent dehydration gives an alpha,beta-unsaturated carbonyl: aldol condensation. Retro-aldol cleavage breaks the alpha–beta C–C bond of a beta-hydroxy carbonyl.

aldol:cue | Memory Cues

Addition adds the partner. Condensation also loses water. The new OH is beta to the retained carbonyl.

aldol:trap | Common Mistakes / MCAT Traps

Aldol products need not contain an aldehyde: ketone donors/acceptors can give beta-hydroxy ketones. Mixed aldols may give mixtures unless donor/acceptor roles are controlled.

aldol:app | MCAT Applications

Aldolase cleavage of carbon skeletons uses this bond logic in metabolism, with enzyme-specific catalysis.

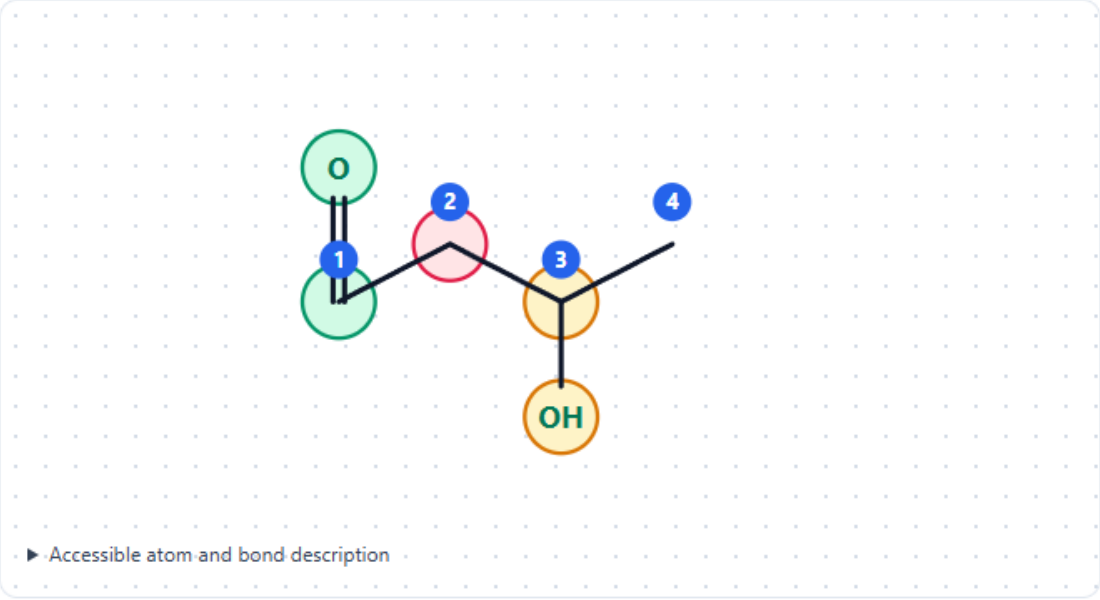
aldol:step-1 | 1. Addition product

Two ethanal units can form 3-hydroxybutanal. The donor retains its carbonyl; the acceptor carbonyl becomes the beta-OH carbon. The newly formed bond joins C2 and C3.

Displayed molecule ID: aldol

{"highlights": {"prin": [0, 4], "alpha": [1], "sub": [2, 5], "num": [0, 1, 2, 3]}, "revealName": true}

Screenshot: aldol-step-1



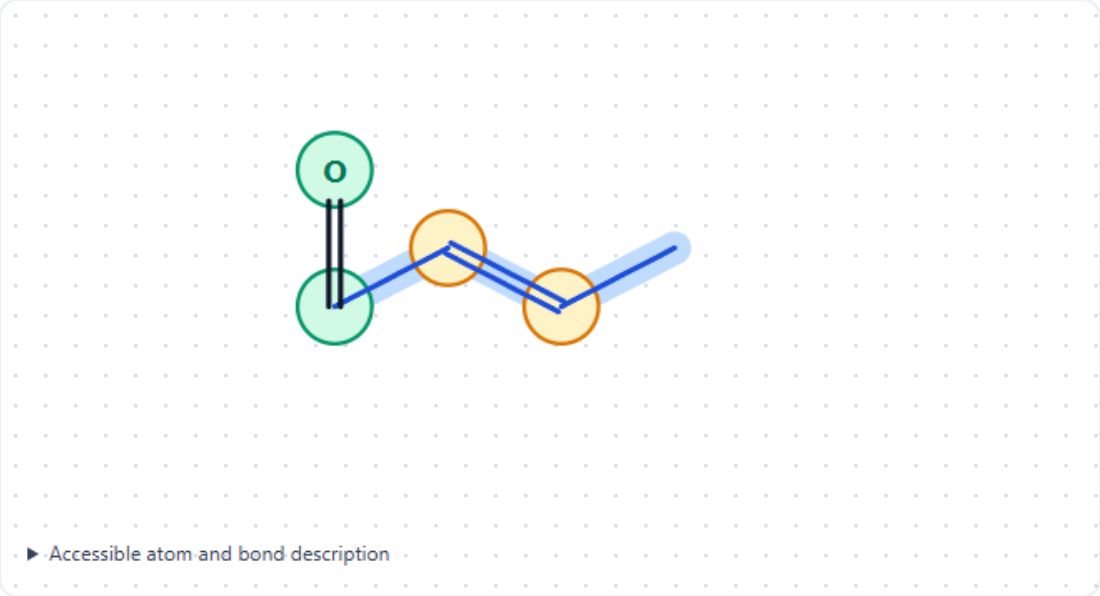
aldol:step-2 | 2. Dehydrate

Loss of water from the alpha/beta region gives a conjugated C=C–C=O system. The diagram specifies connectivity only, not an E/Z assignment.

Displayed molecule ID: crotonaldehyde

{"highlights": {"path": [0, 1, 2, 3], "prin": [0, 4], "sub": [1, 2]}, "revealName": true}

Screenshot: aldol-step-2



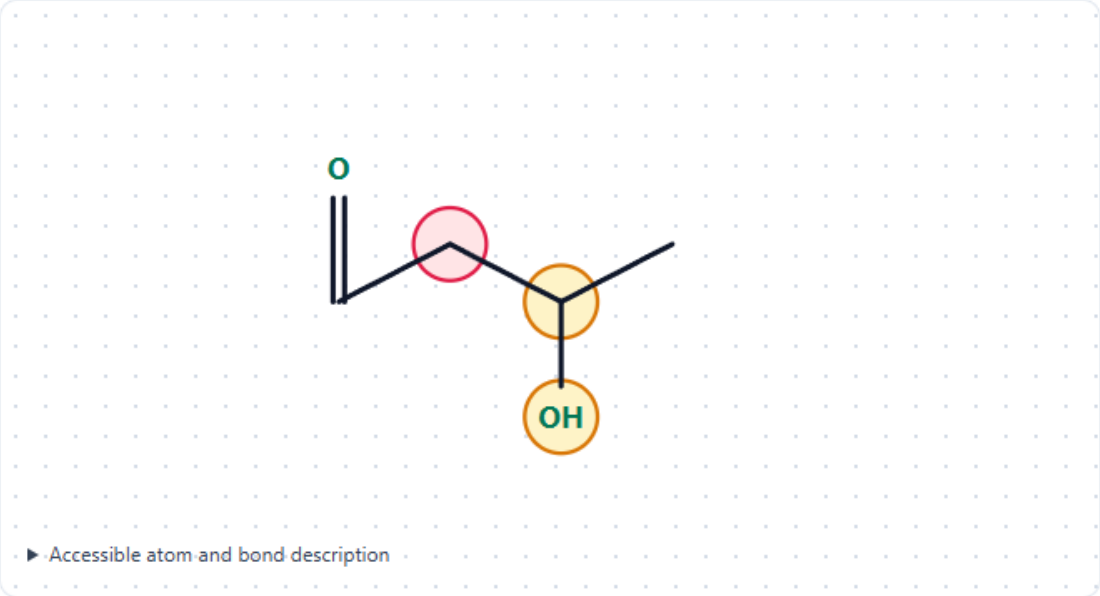
aldol:step-3 | 3. Reverse the carbon–carbon step

In a retro-aldol, cleave the bond between alpha and beta carbons of the beta-hydroxy carbonyl; regenerate carbonyl-containing fragments.

Displayed molecule ID: aldol

{"highlights": {"alpha": [1], "sub": [2, 5]}, "revealName": false}

Screenshot: aldol-step-3



conjugate | Conjugate addition and enamine nucleophiles

An enone has more than one electrophilic site

conjugate:rule | Core Rules

An alpha,beta-unsaturated carbonyl can undergo direct 1,2-addition at carbonyl carbon or conjugate 1,4-addition at beta carbon. Enolates and enamines can serve as carbon nucleophiles; their selectivity depends on reagent and conditions.

conjugate:cue | Memory Cues

Michael addition: nucleophile bonds to beta carbon, then protonation/tautomerization restores the carbonyl.

conjugate:trap | Common Mistakes / MCAT Traps

Do not label every nucleophile addition to an enone “Michael.” Direct carbonyl attack is 1,2-addition. Enamines are neutral nitrogen analogues used for alpha functionalization, then hydrolyzed back to carbonyls.

conjugate:app | MCAT Applications

Conjugate acceptors can react with biological nucleophiles; the site of reaction matters for covalent inhibitor design.

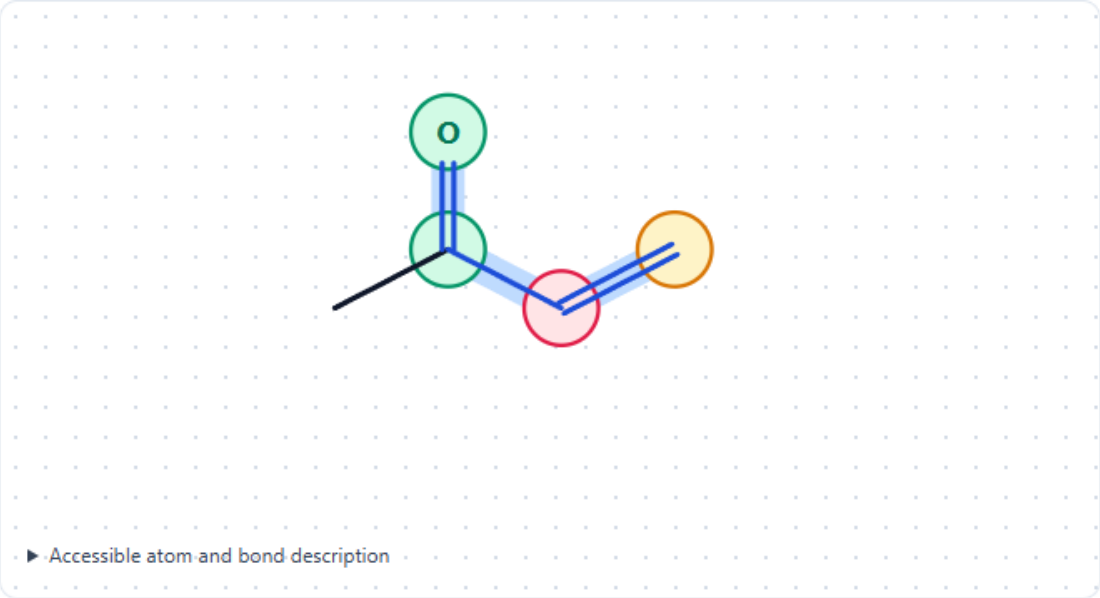
conjugate:step-1 | 1. Locate the conjugated system

The carbonyl carbon is B, alpha carbon is C, and beta carbon is D. Alternating pi/sigma bonding allows electron redistribution.

Displayed molecule ID: methylvinylketone

{"highlights": {"path": [4, 1, 2, 3], "prin": [1, 4], "alpha": [2], "sub": [3]}, "revealName": true}

Screenshot: conjugate-step-1



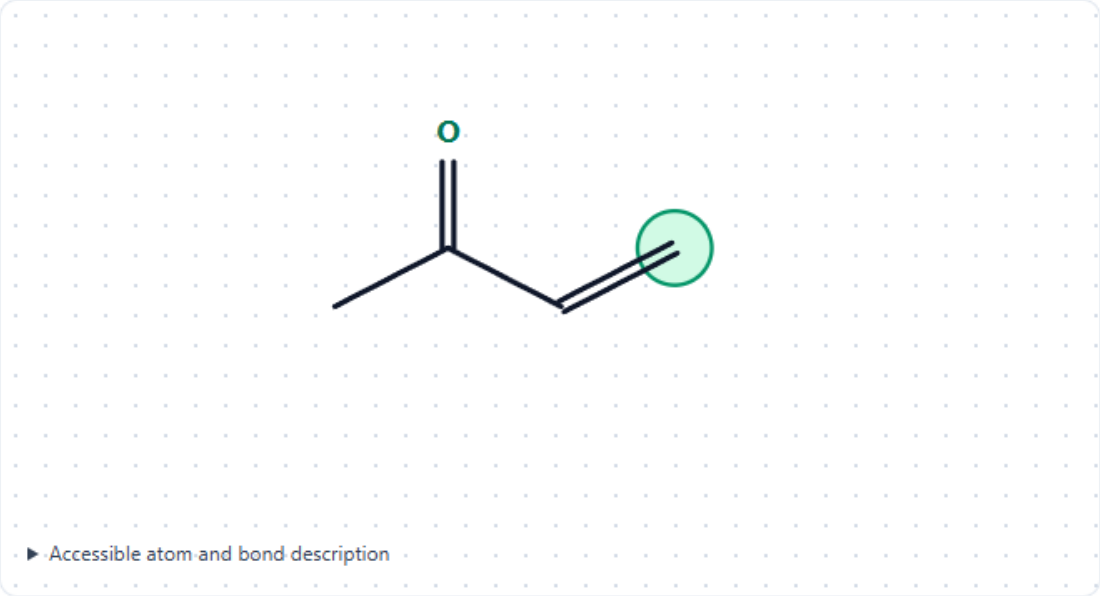
conjugate:step-2 | 2. Choose the beta site

A Michael-type conjugate addition makes a new bond at the beta carbon. The carbonyl is retained after the enolate-like intermediate is protonated.

Displayed molecule ID: methylvinylketone

{"highlights": {"prin": [3]}, "revealName": false}

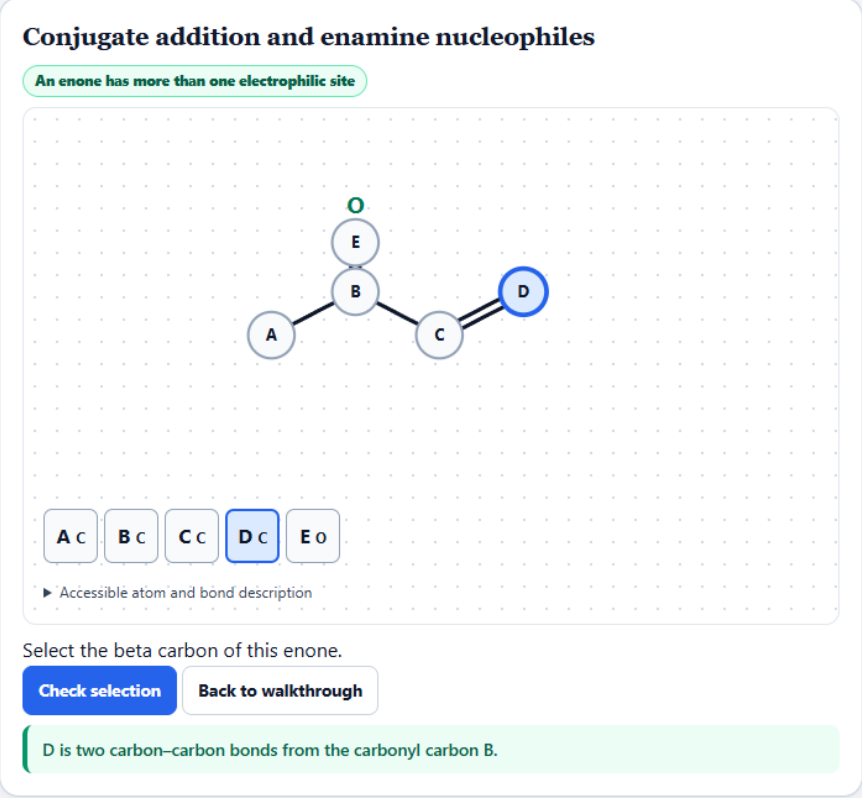
Screenshot: conjugate-step-2



Embedded challenge: Select the beta carbon of this enone.

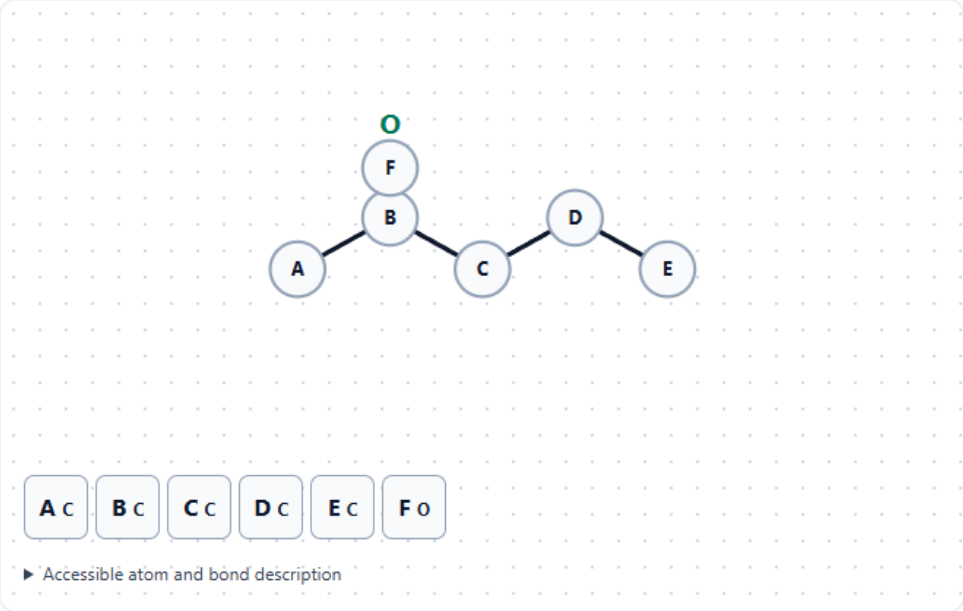
{"prompt": "Select the beta carbon of this enone.", "answer": [3], "explanation": "D is two carbon–carbon bonds from the carbonyl carbon B."}

Screenshot: conjugate-step-2-challenge



alpha-map | Identify enolate-forming sites

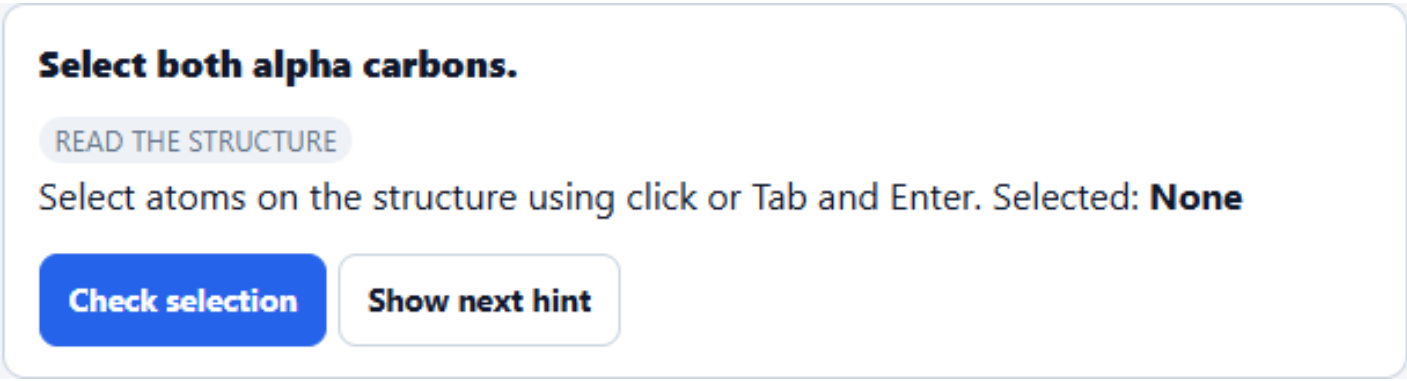
Screenshot: alpha-map-stage-1-molecule



alpha-map:stage-1 | Select both alpha carbons.

Answer type: atoms

Screenshot: alpha-map-stage-1-question



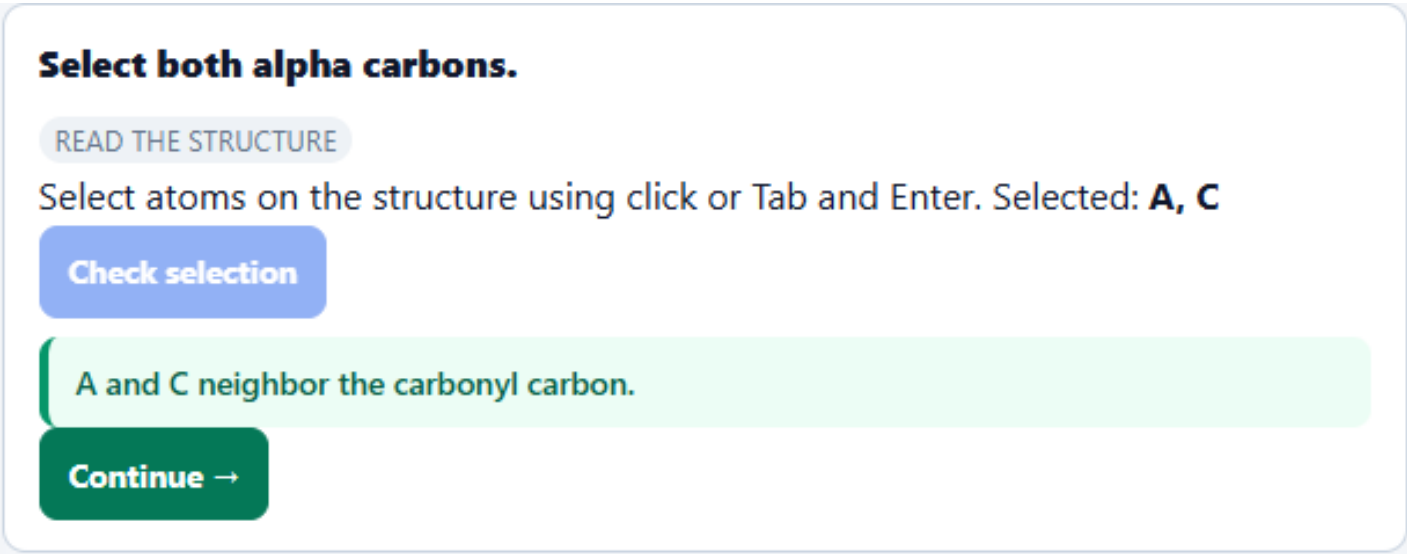
{"authorAnswer": [0, 2], "tolerance": null, "highlightsAfterCorrectAnswer": {"alpha": [0, 2]}}

Hint 1: Locate C=O first.

Hint 2: Choose carbon neighbors of that carbon.

Explanation: A and C neighbor the carbonyl carbon.

Screenshot: alpha-map-stage-1-feedback



alpha-map:stage-2 | Why are their hydrogens relatively acidic?

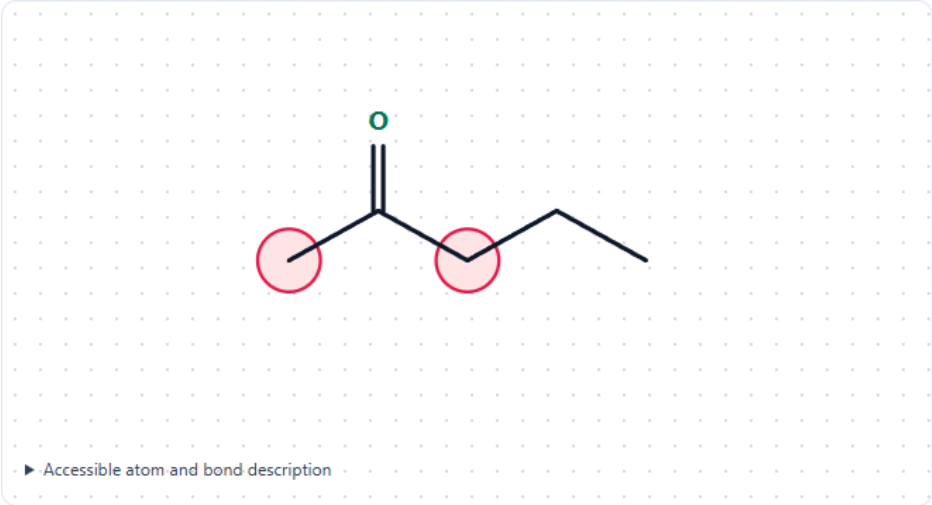
Answer type: choice

Option 0 (A): The conjugate base is resonance stabilized

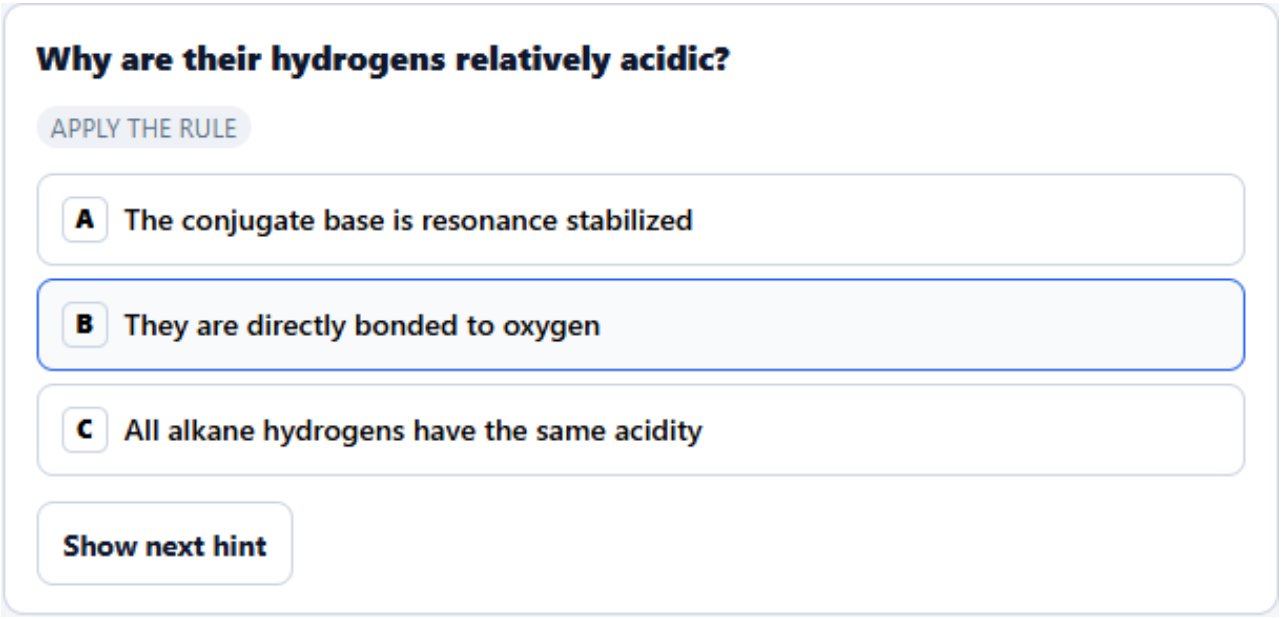
Option 1 (B): They are directly bonded to oxygen

Option 2 (C): All alkane hydrogens have the same acidity

Screenshot: alpha-map-stage-2-molecule



Screenshot: alpha-map-stage-2-question



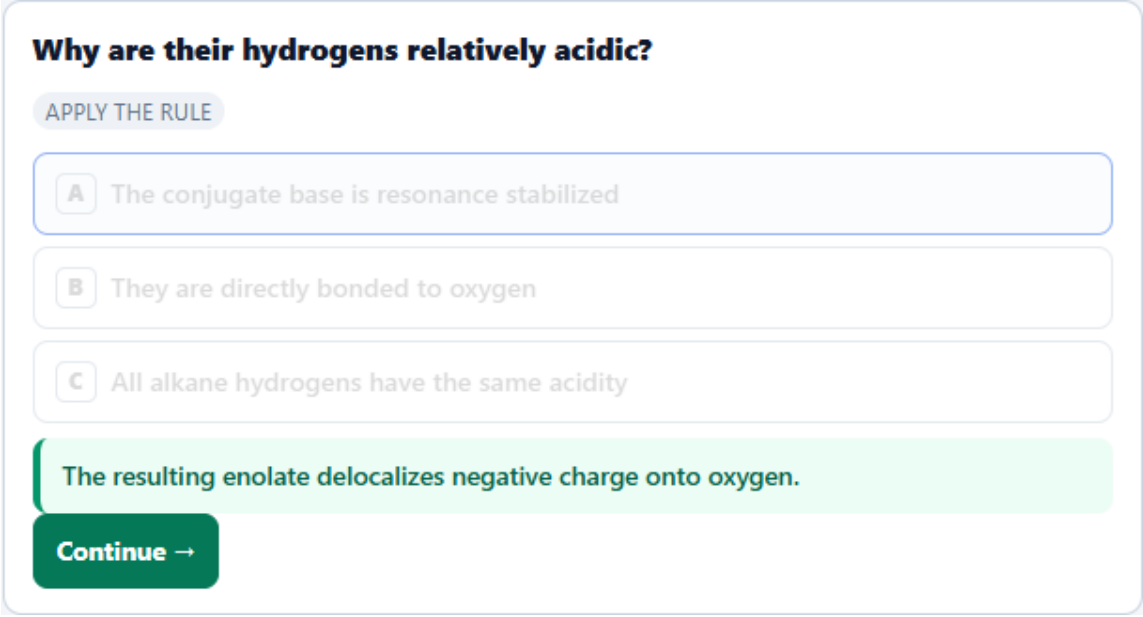
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Compare the stability of the conjugate base.

Hint 2: Adjacent p orbitals enable delocalization.

Explanation: The resulting enolate delocalizes negative charge onto oxygen.

Screenshot: alpha-map-stage-2-feedback



alpha-map:stage-3 | Enolate resonance contributors differ in…

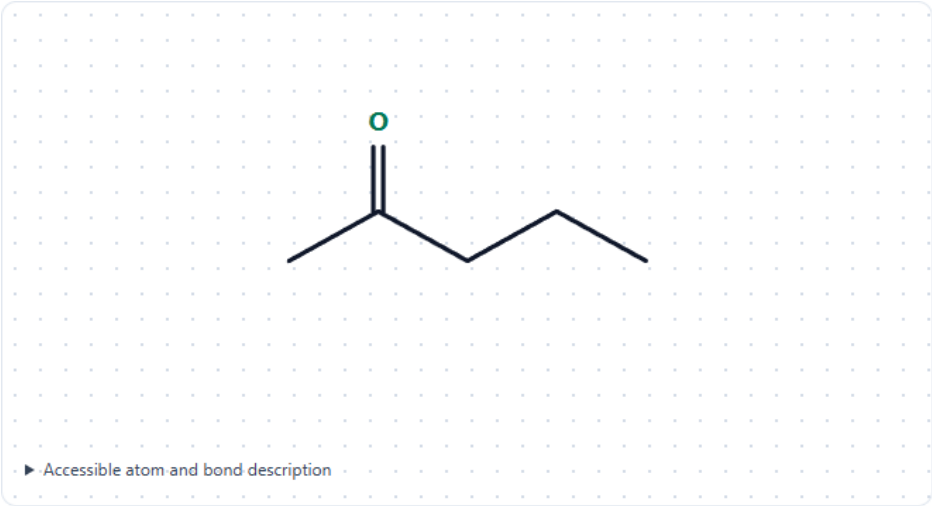
Answer type: choice

Option 0 (A): Electron placement

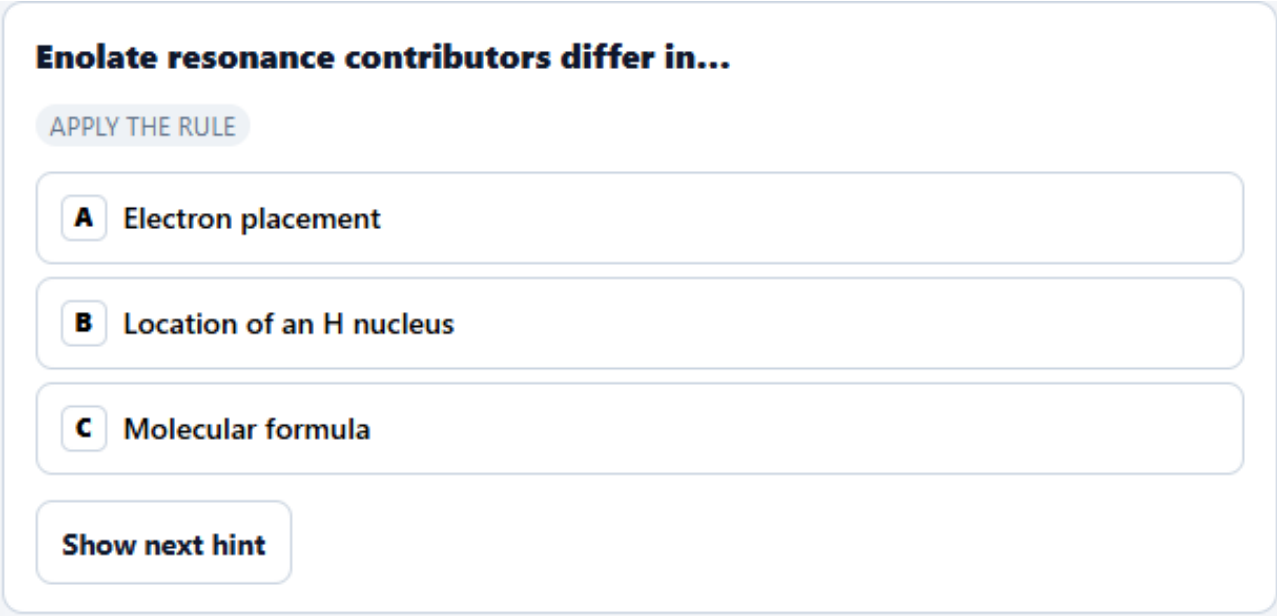
Option 1 (B): Location of an H nucleus

Option 2 (C): Molecular formula

Screenshot: alpha-map-stage-3-molecule



Screenshot: alpha-map-stage-3-question



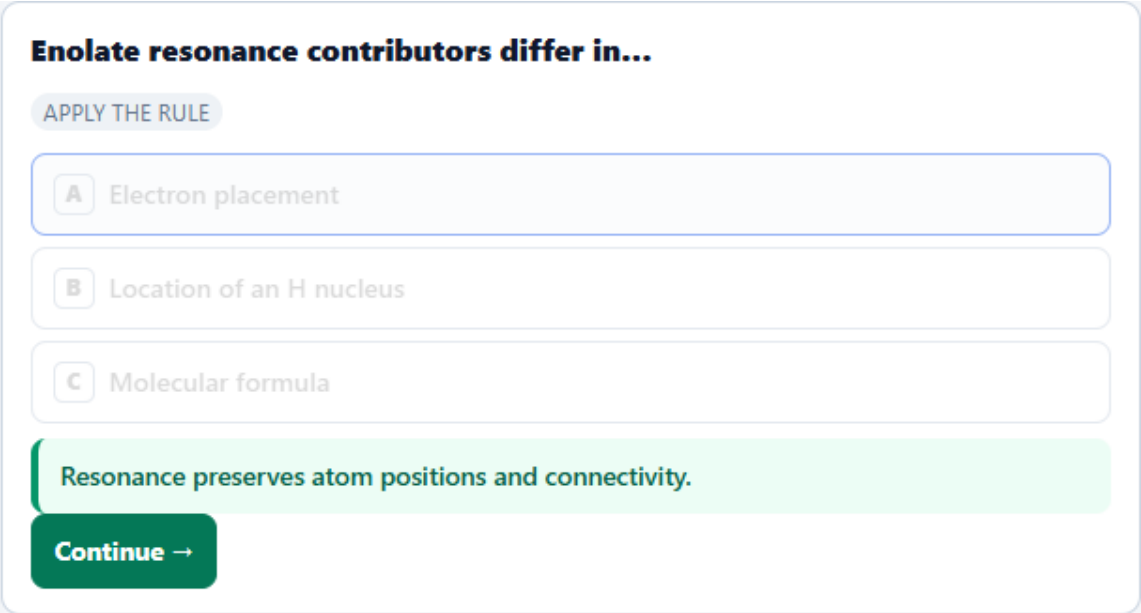
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Distinguish resonance from tautomerization.

Hint 2: No atoms move between contributors.

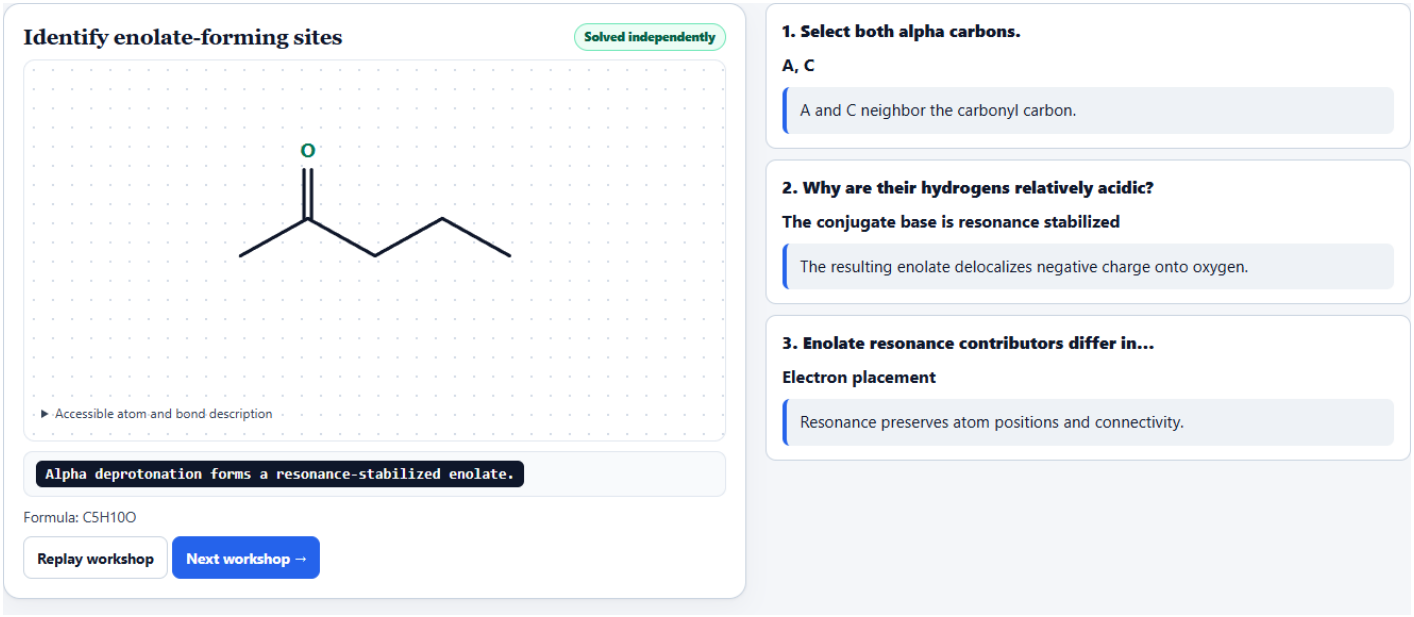
Explanation: Resonance preserves atom positions and connectivity.

Screenshot: alpha-map-stage-3-feedback



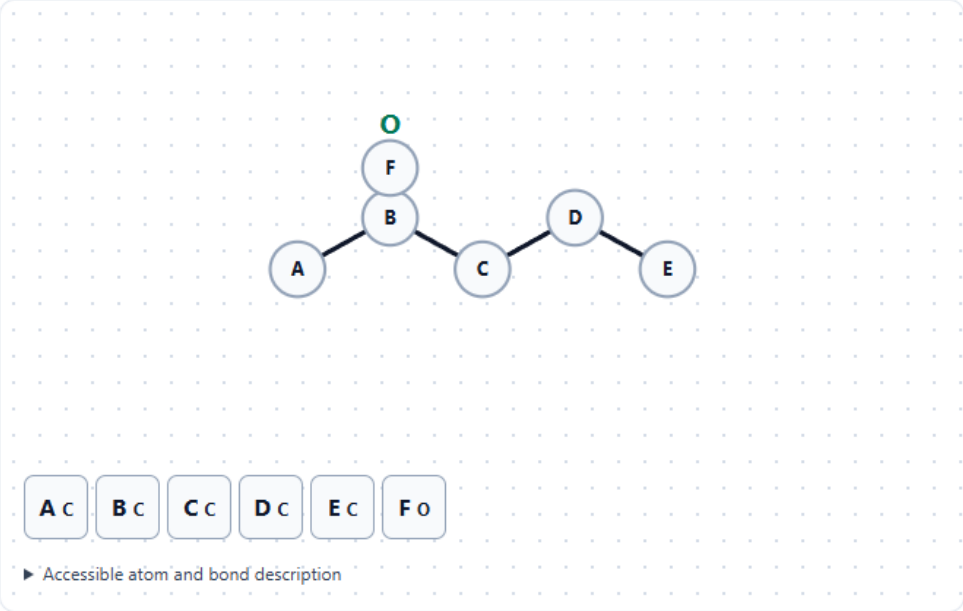
Completion summary: Alpha deprotonation forms a resonance-stabilized enolate.

Screenshot: alpha-map-complete



enolate-control | Choose kinetic or equilibrium control

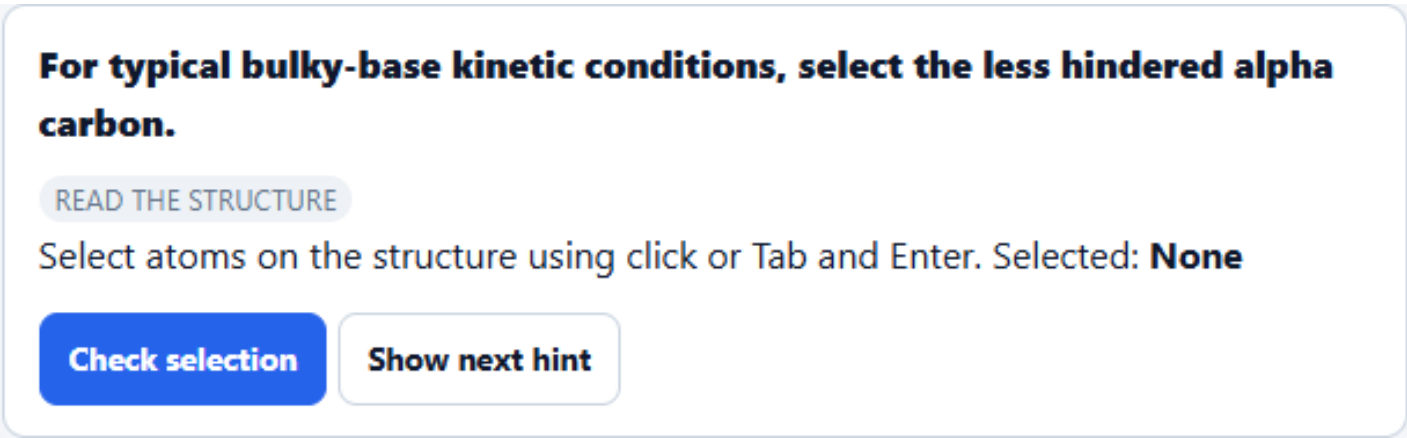
Screenshot: enolate-control-stage-1-molecule



enolate-control:stage-1 | For typical bulky-base kinetic conditions, select the less hindered alpha carbon.

Answer type: atoms

Screenshot: enolate-control-stage-1-question



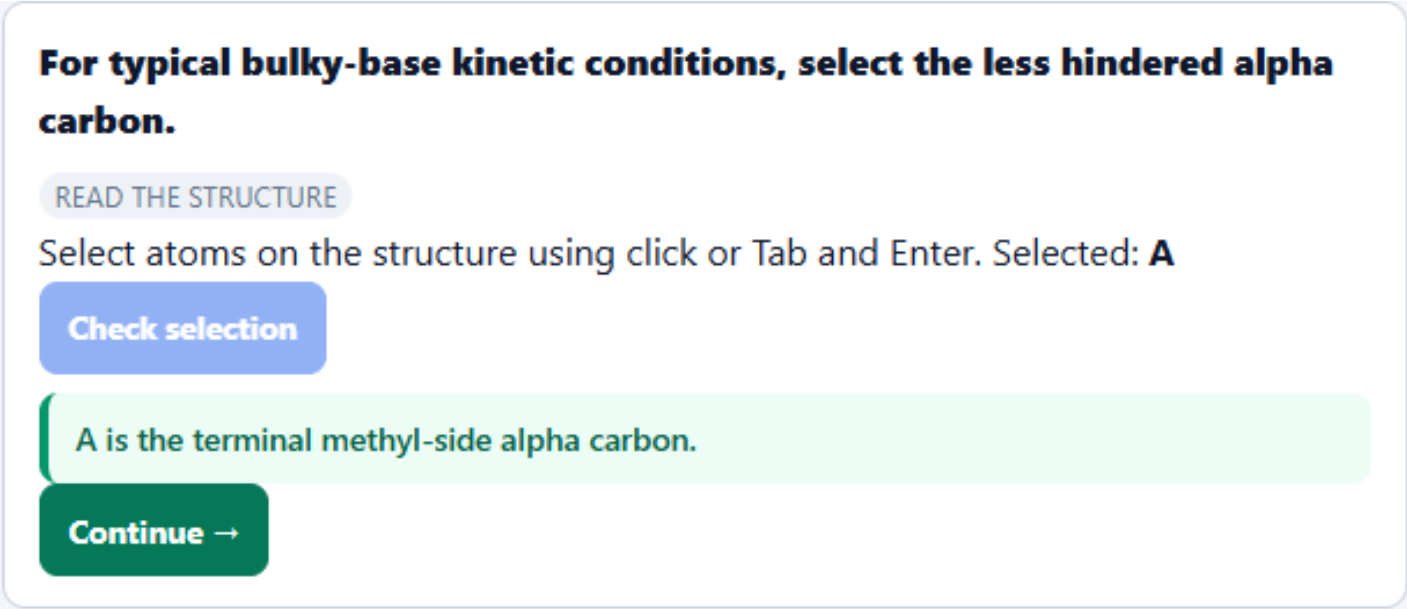
{"authorAnswer": [0], "tolerance": null, "highlightsAfterCorrectAnswer": {"alpha": [0]}}

Hint 1: Compare the two alpha sites.

Hint 2: The terminal methyl side is less substituted.

Explanation: A is the terminal methyl-side alpha carbon.

Screenshot: enolate-control-stage-1-feedback



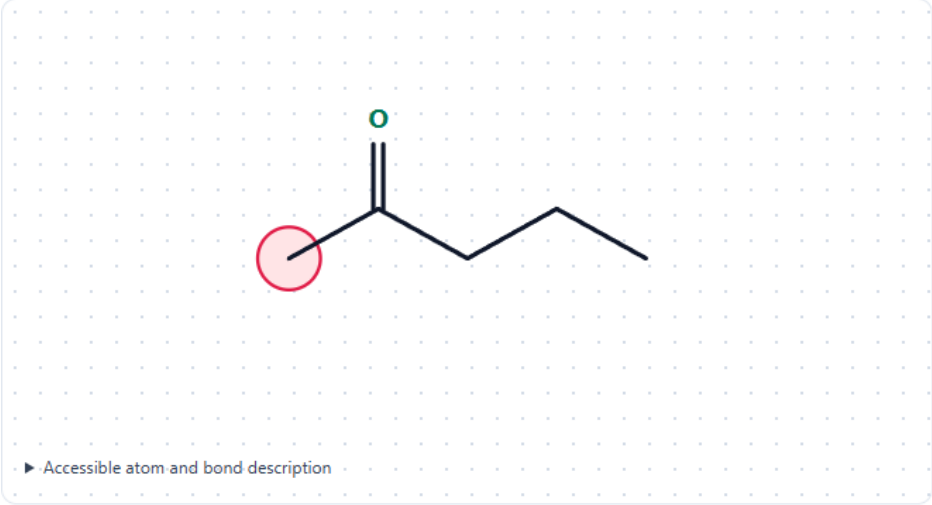
enolate-control:stage-2 | Conditions favoring kinetic enolate formation are usually…

Answer type: choice

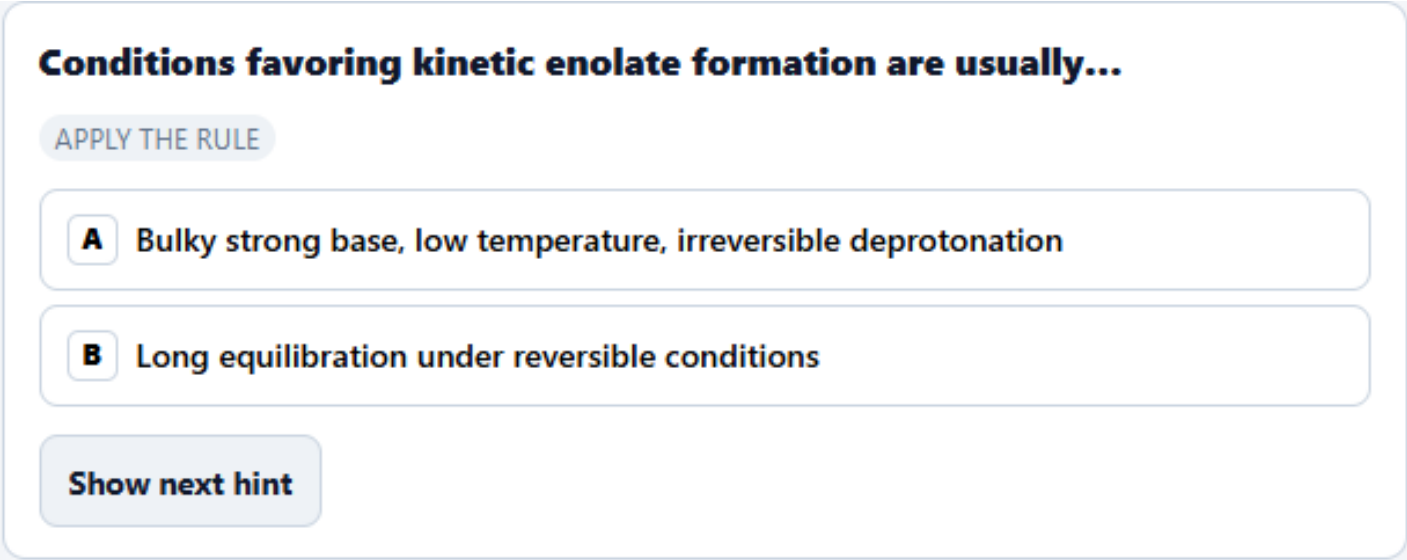
Option 0 (A): Bulky strong base, low temperature, irreversible deprotonation

Option 1 (B): Long equilibration under reversible conditions

Screenshot: enolate-control-stage-2-molecule



Screenshot: enolate-control-stage-2-question



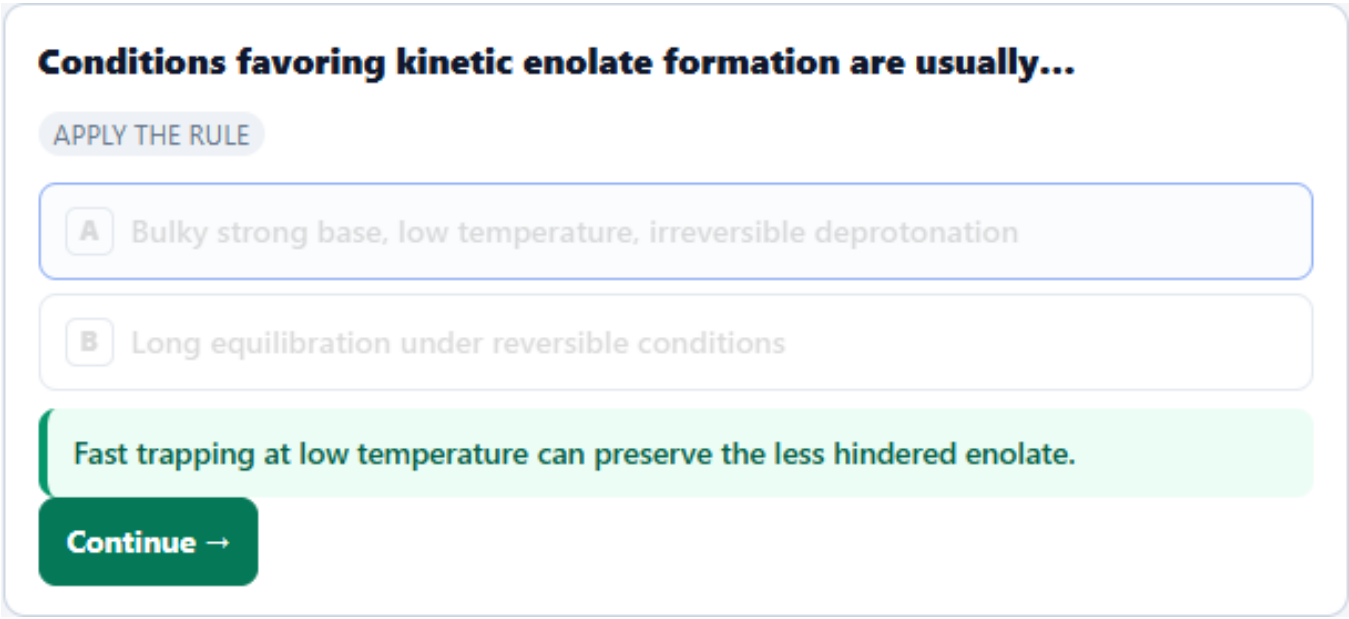
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Kinetic control tracks the lower activation barrier.

Hint 2: Avoid allowing products to equilibrate.

Explanation: Fast trapping at low temperature can preserve the less hindered enolate.

Screenshot: enolate-control-stage-2-feedback



enolate-control:stage-3 | Neutral keto and enol forms are…

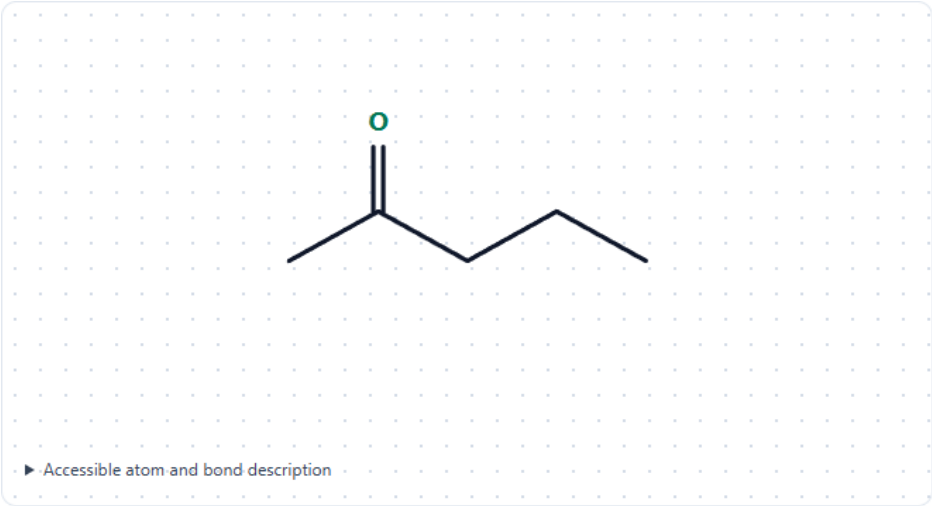
Answer type: choice

Option 0 (A): Tautomers

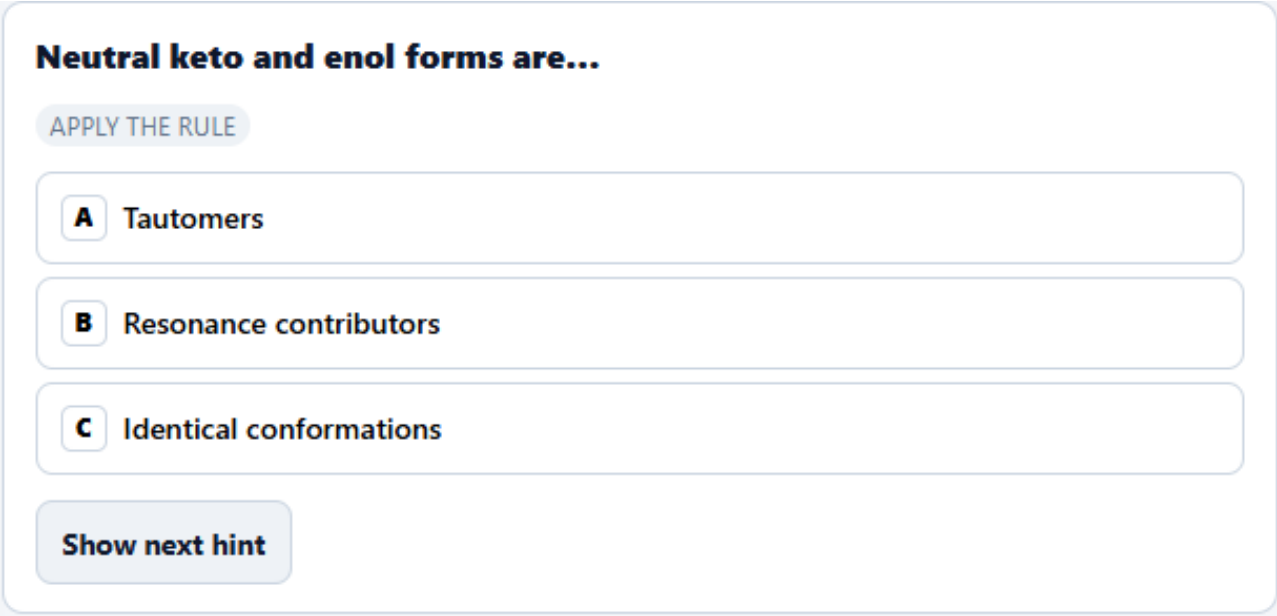
Option 1 (B): Resonance contributors

Option 2 (C): Identical conformations

Screenshot: enolate-control-stage-3-molecule



Screenshot: enolate-control-stage-3-question



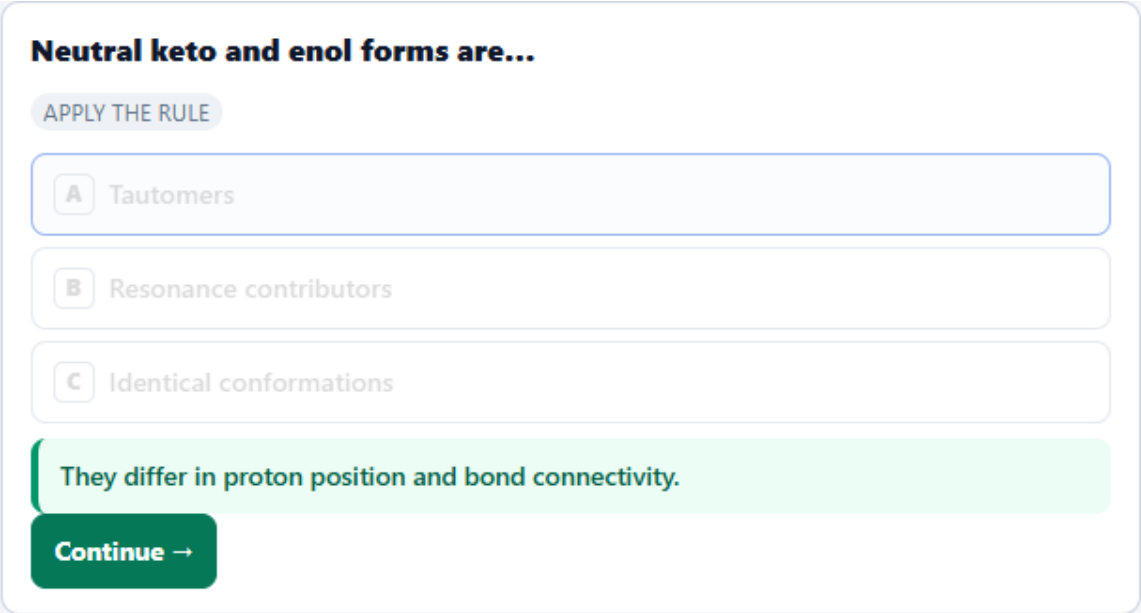
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Check whether an H nucleus moves.

Hint 2: Resonance cannot move nuclei.

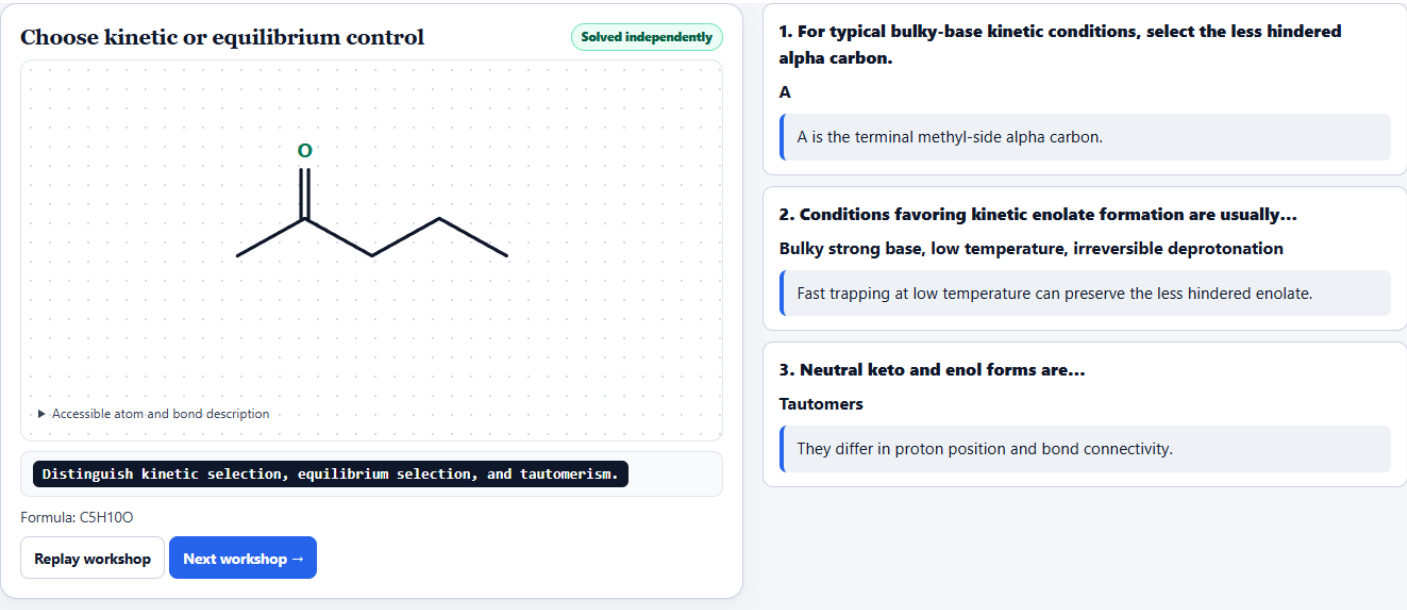
Explanation: They differ in proton position and bond connectivity.

Screenshot: enolate-control-stage-3-feedback



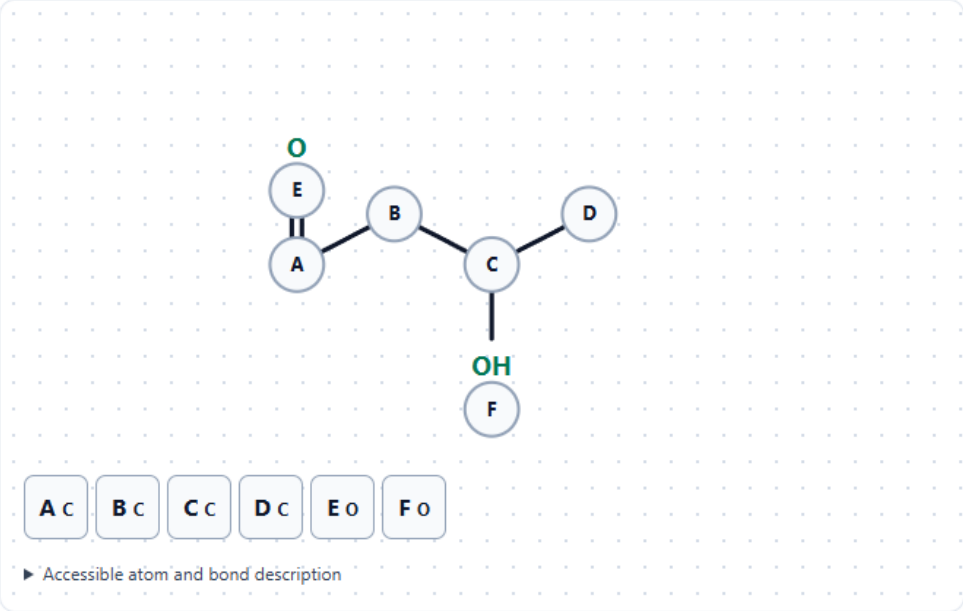
Completion summary: Distinguish kinetic selection, equilibrium selection, and tautomerism.

Screenshot: enolate-control-complete



aldol-map | Trace an aldol product

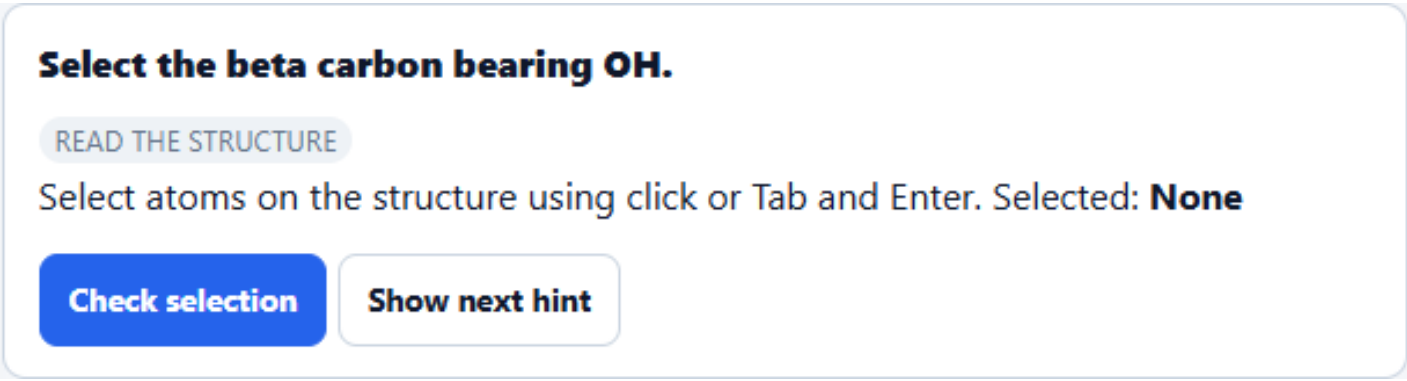
Screenshot: aldol-map-stage-1-molecule



aldol-map:stage-1 | Select the beta carbon bearing OH.

Answer type: atoms

Screenshot: aldol-map-stage-1-question



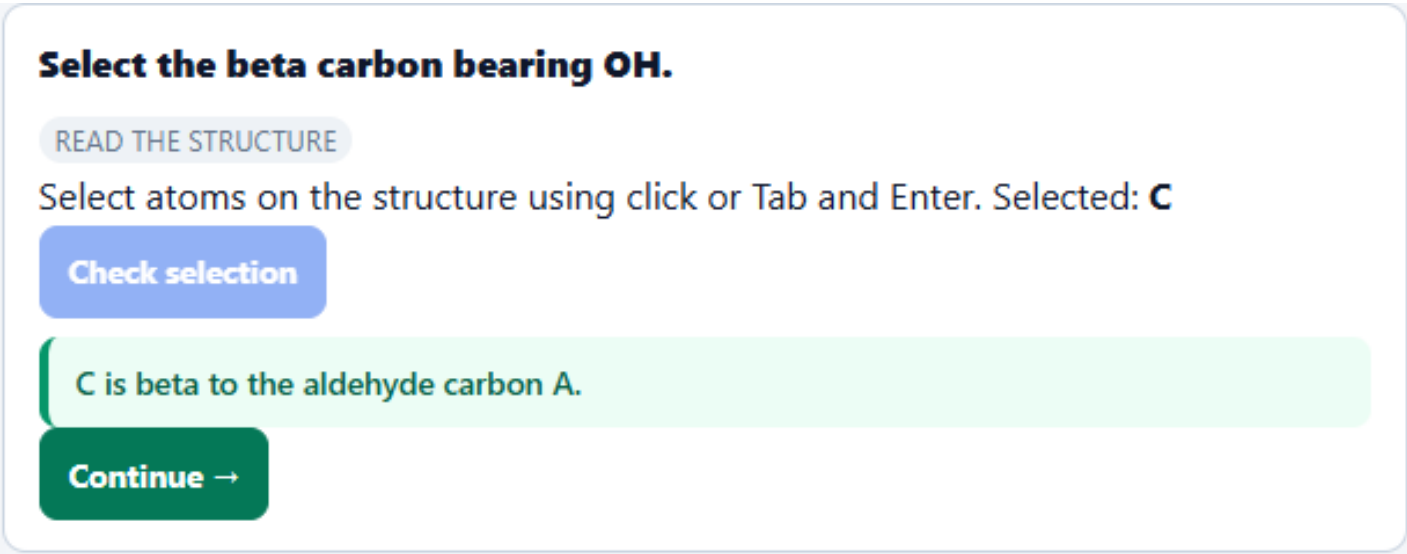
{"authorAnswer": [2], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [0, 4], "alpha": [1], "sub": [2, 5]}}

Hint 1: Start at carbonyl carbon A.

Hint 2: B is alpha; the next carbon is beta.

Explanation: C is beta to the aldehyde carbon A.

Screenshot: aldol-map-stage-1-feedback



aldol-map:stage-2 | The displayed beta-hydroxy product is formed by…

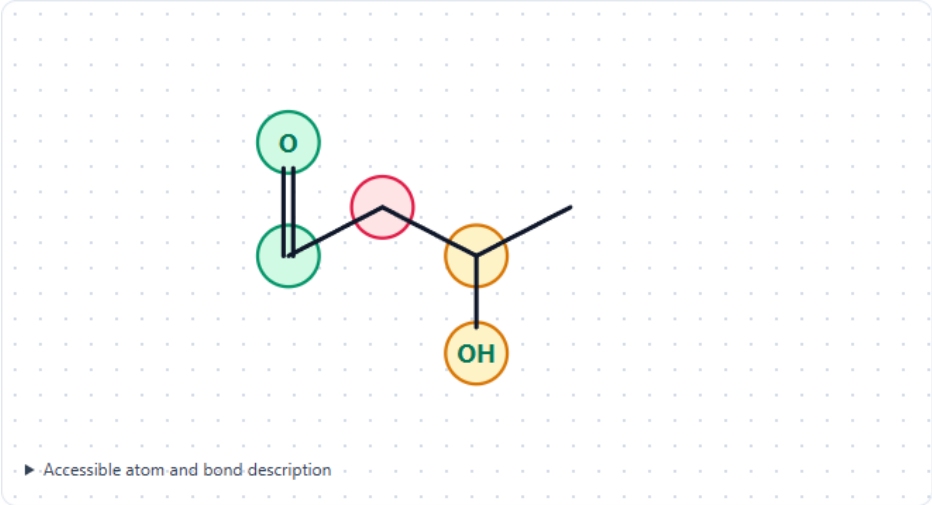
Answer type: choice

Option 0 (A): Aldol addition

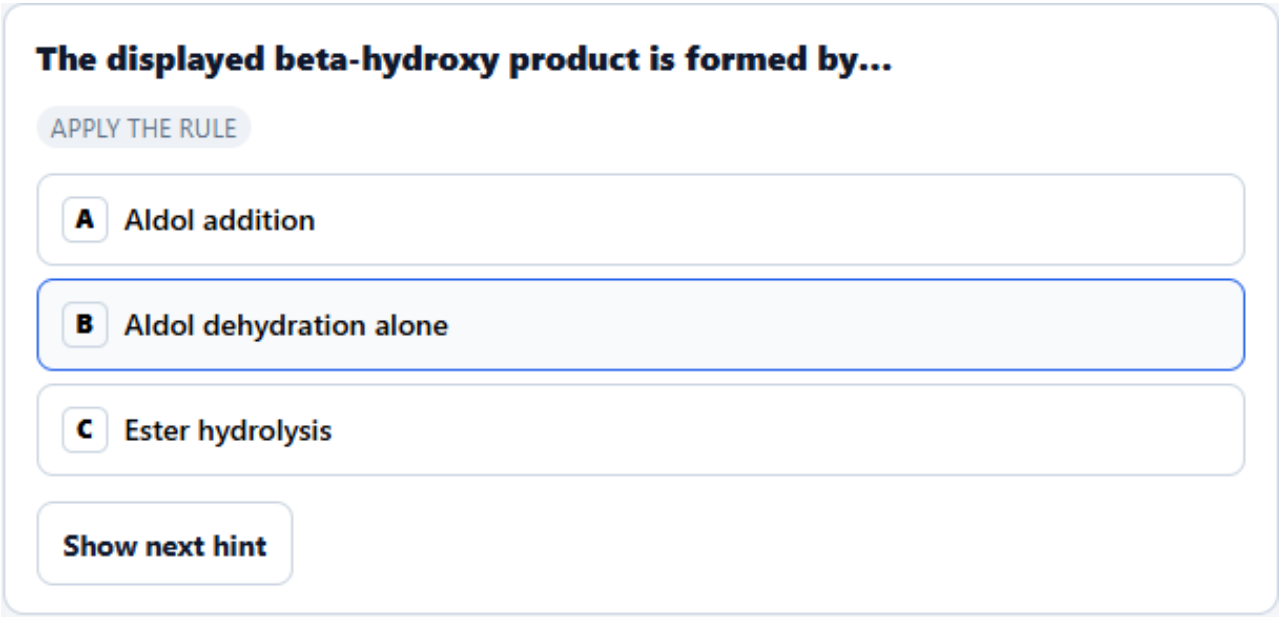
Option 1 (B): Aldol dehydration alone

Option 2 (C): Ester hydrolysis

Screenshot: aldol-map-stage-2-molecule



Screenshot: aldol-map-stage-2-question



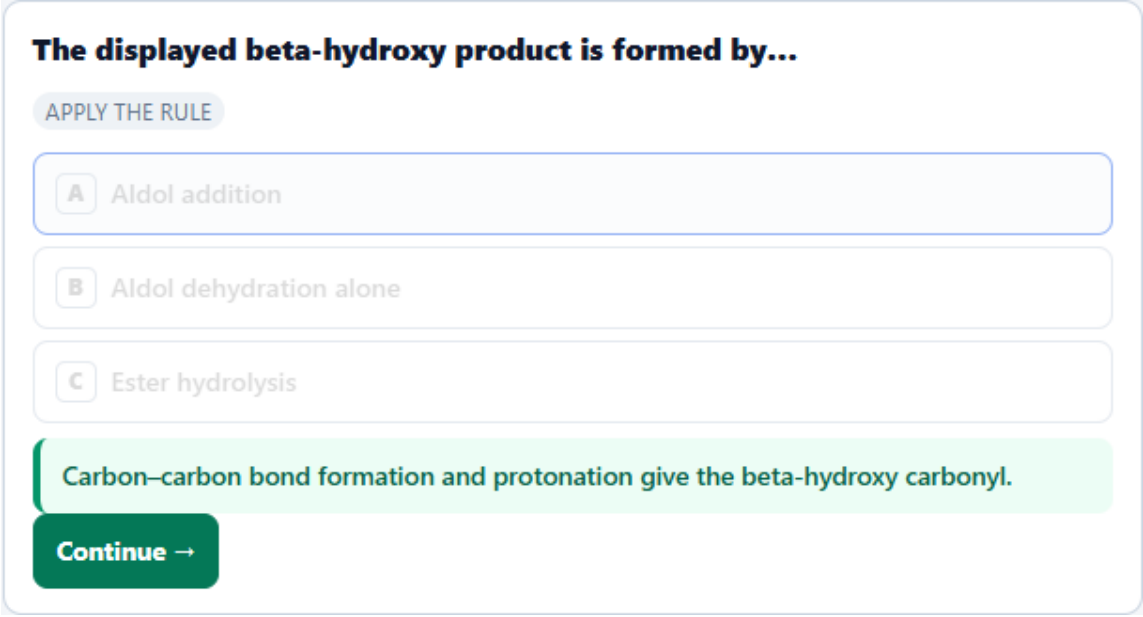
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: The OH is still present.

Hint 2: Water loss has not yet generated a conjugated alkene.

Explanation: Carbon–carbon bond formation and protonation give the beta-hydroxy carbonyl.

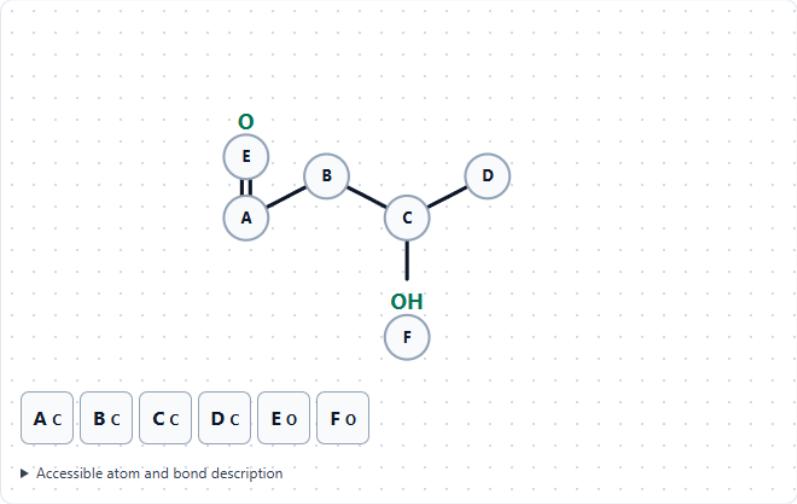
Screenshot: aldol-map-stage-2-feedback



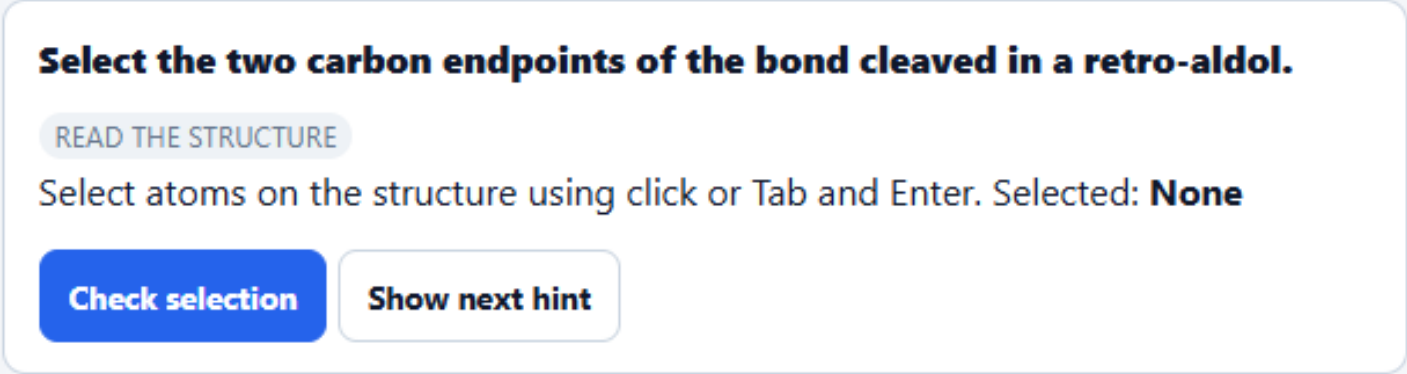
aldol-map:stage-3 | Select the two carbon endpoints of the bond cleaved in a retro-aldol.

Answer type: atoms

Screenshot: aldol-map-stage-3-molecule



Screenshot: aldol-map-stage-3-question



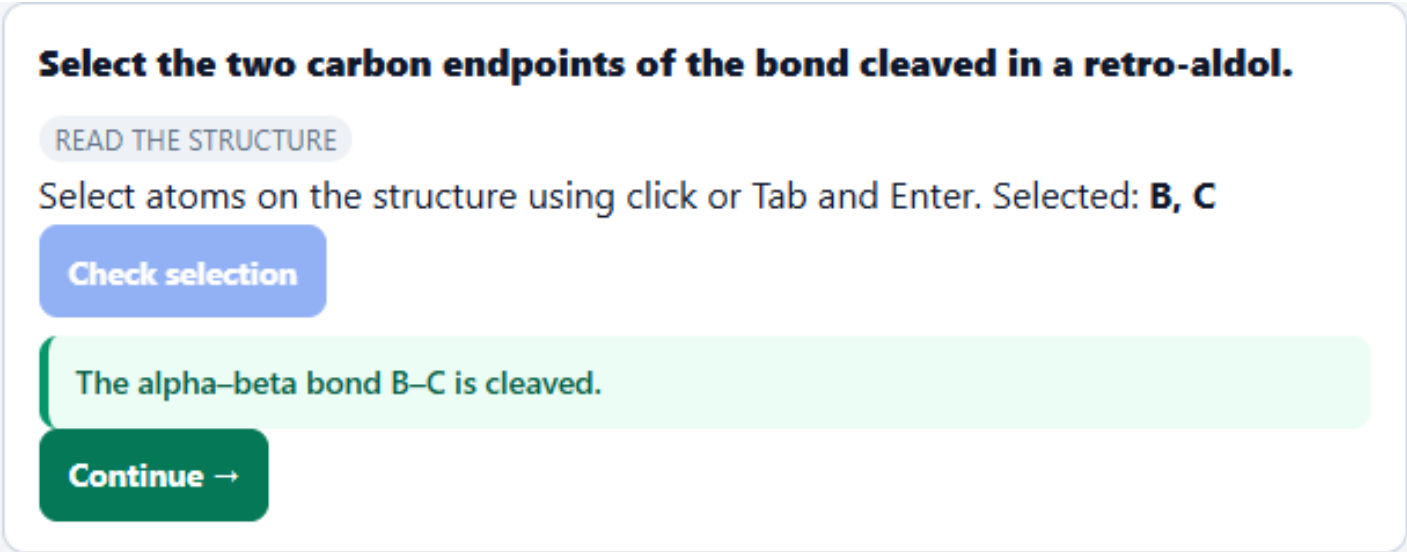
{"authorAnswer": [1, 2], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 2]}}

Hint 1: Reverse the new carbon–carbon bond formation.

Hint 2: Connect the alpha carbon to the OH-bearing beta carbon.

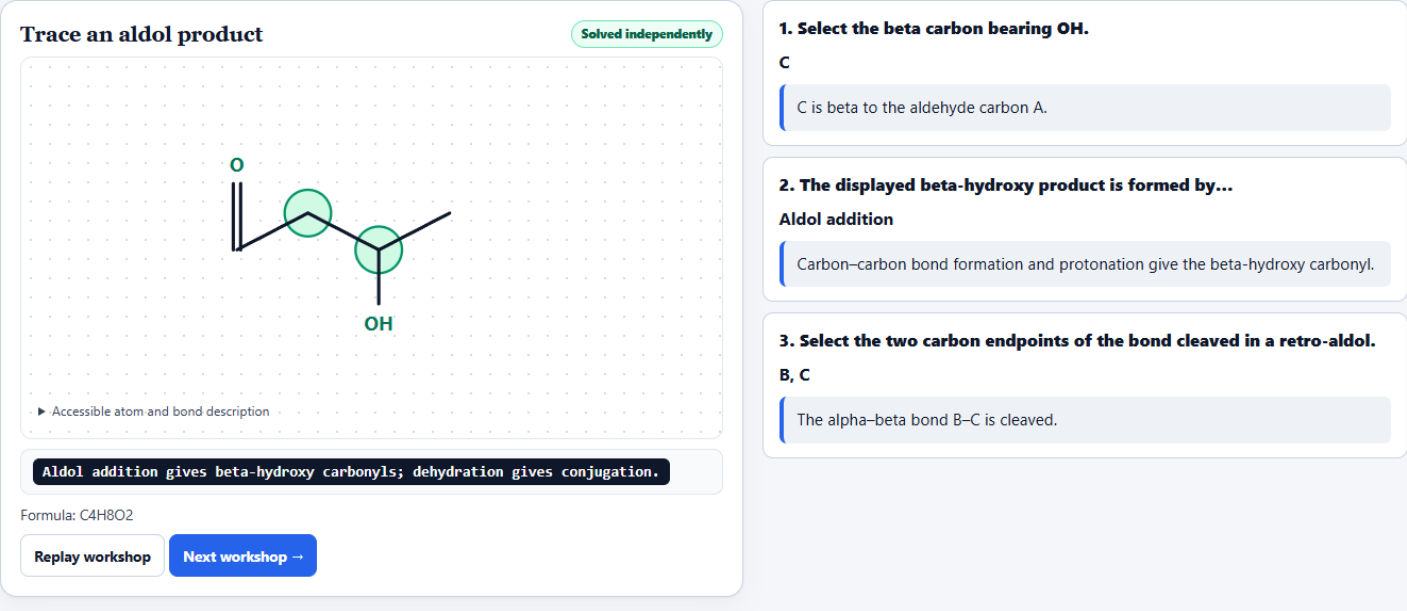
Explanation: The alpha–beta bond B–C is cleaved.

Screenshot: aldol-map-stage-3-feedback



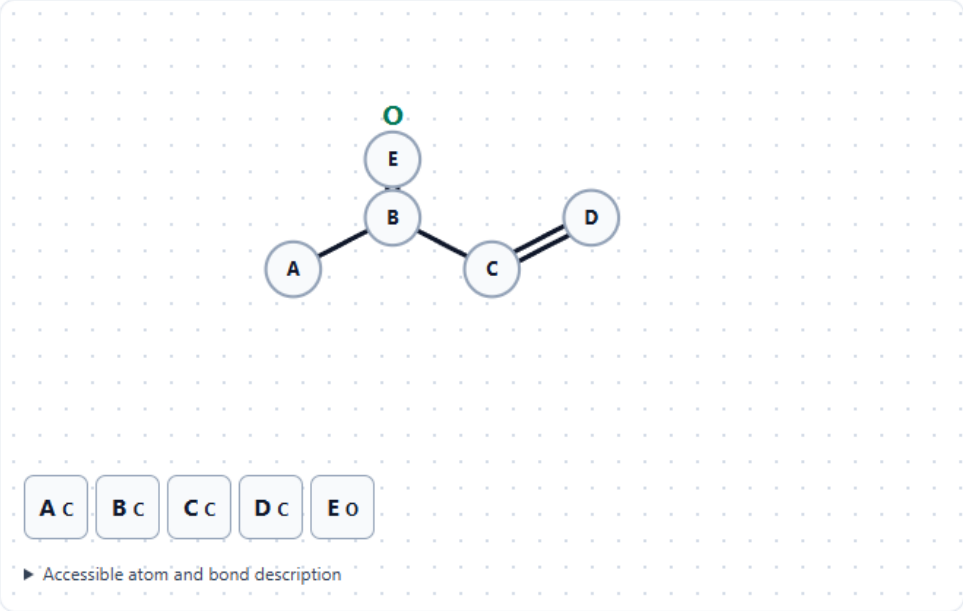
Completion summary: Aldol addition gives beta-hydroxy carbonyls; dehydration gives conjugation.

Screenshot: aldol-map-complete



michael-site | Find a conjugate-addition site

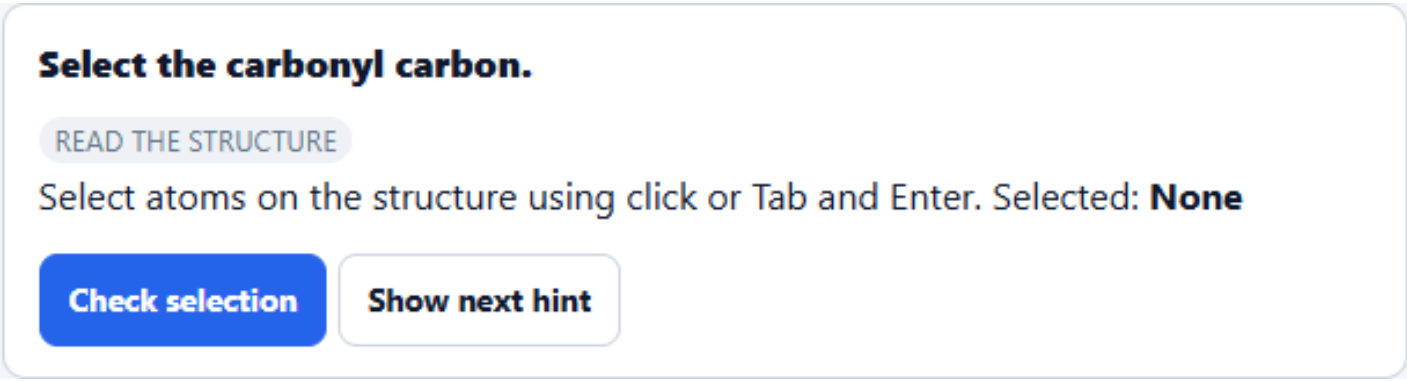
Screenshot: michael-site-stage-1-molecule



michael-site:stage-1 | Select the carbonyl carbon.

Answer type: atoms

Screenshot: michael-site-stage-1-question



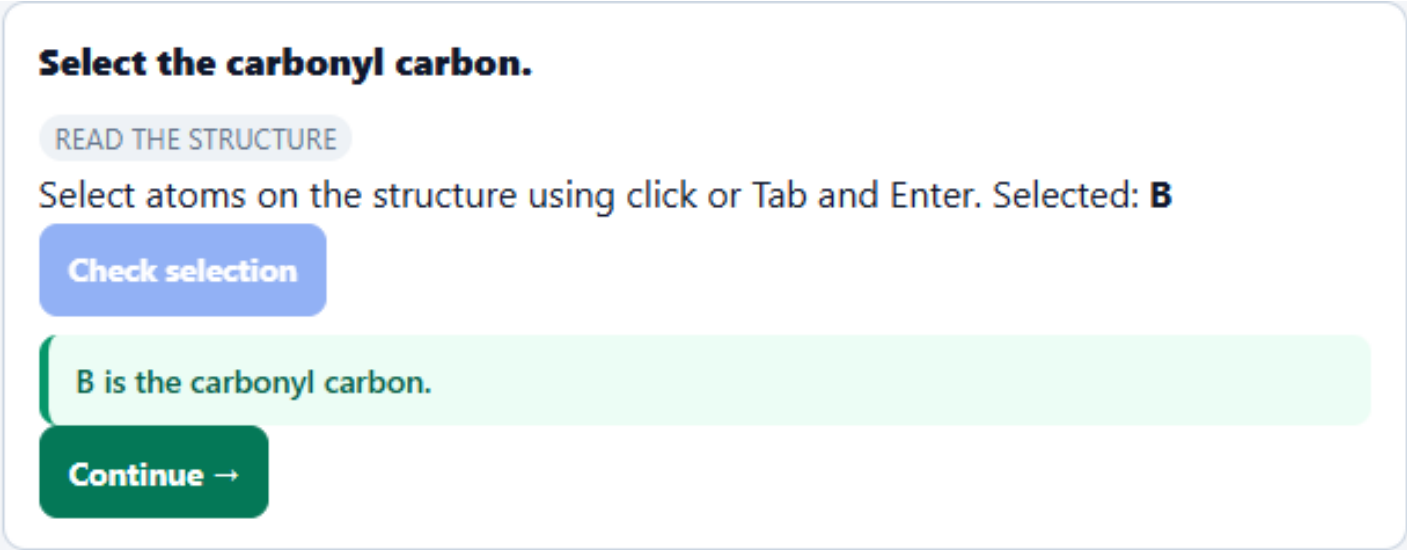
{"authorAnswer": [1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 4]}}

Hint 1: Find C=O.

Hint 2: Choose the carbon endpoint.

Explanation: B is the carbonyl carbon.

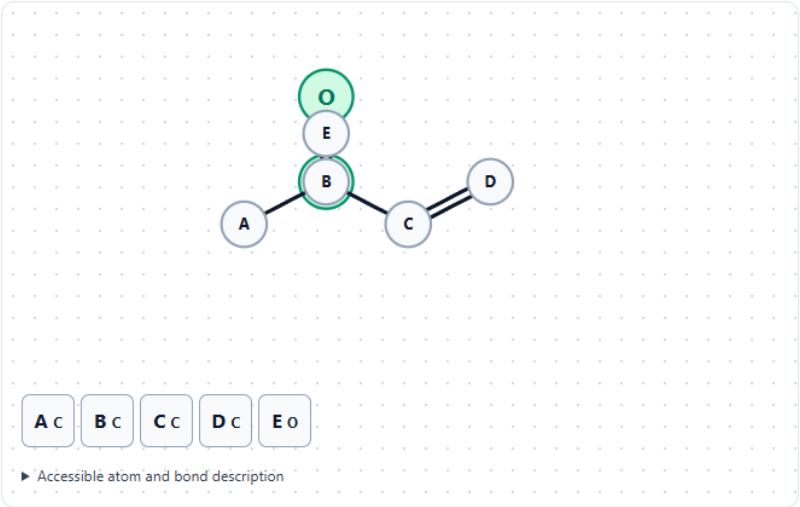
Screenshot: michael-site-stage-1-feedback



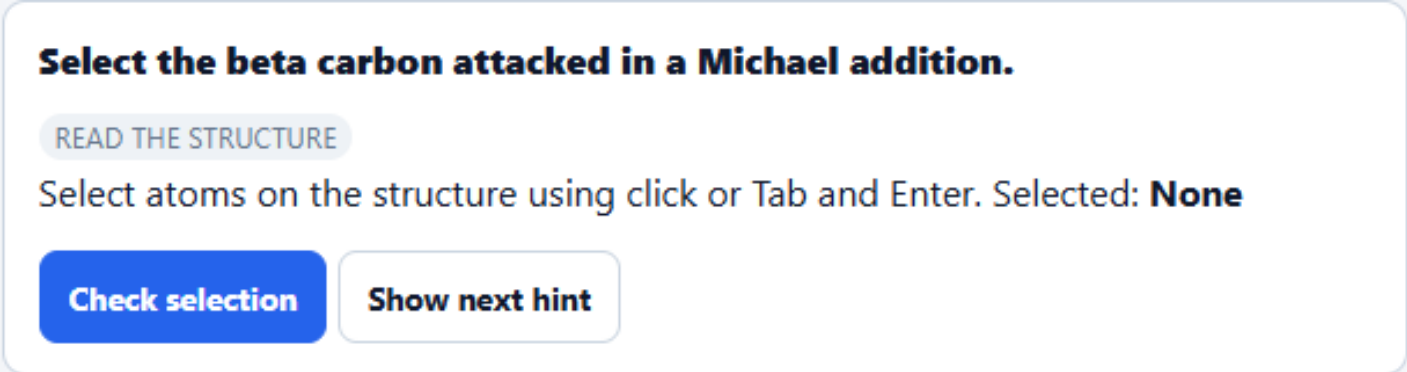
michael-site:stage-2 | Select the beta carbon attacked in a Michael addition.

Answer type: atoms

Screenshot: michael-site-stage-2-molecule



Screenshot: michael-site-stage-2-question



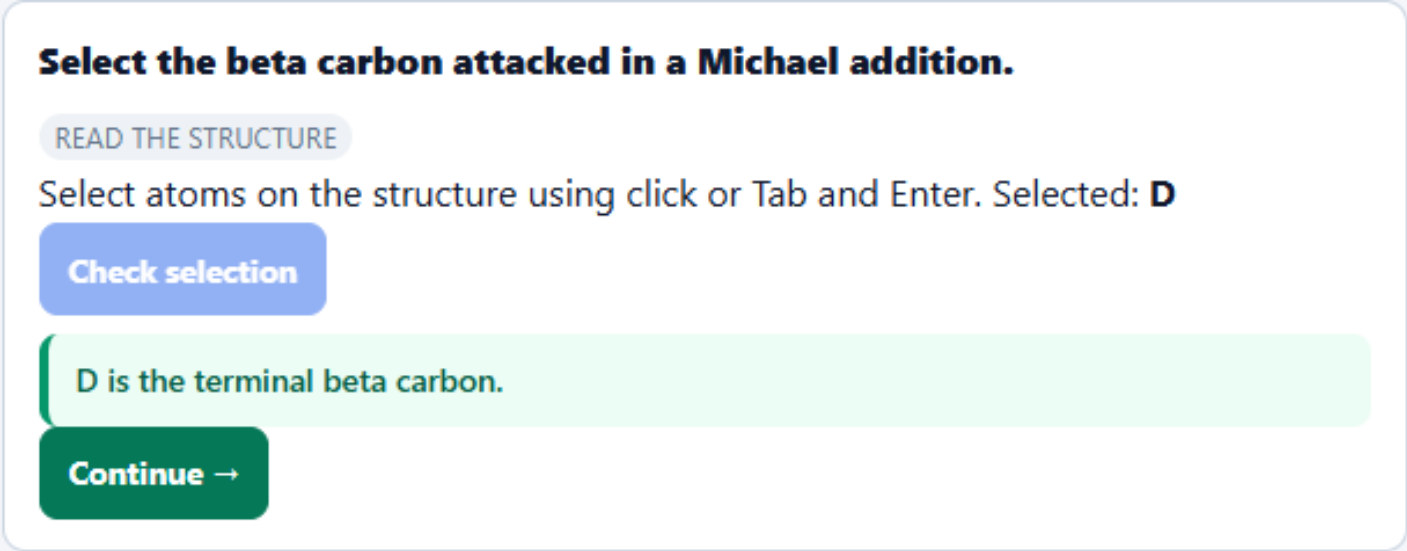
{"authorAnswer": [3], "tolerance": null, "highlightsAfterCorrectAnswer": {"sub": [3]}}

Hint 1: Move one carbon to alpha, then another to beta.

Hint 2: The alkene extends away from C=O.

Explanation: D is the terminal beta carbon.

Screenshot: michael-site-stage-2-feedback



michael-site:stage-3 | Attack directly at the carbonyl carbon is called…

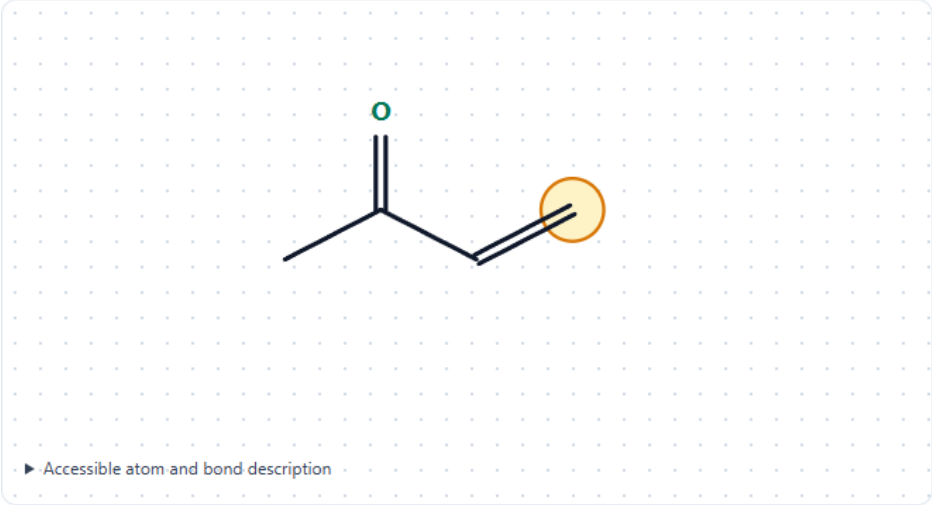
Answer type: choice

Option 0 (A): 1,2-addition

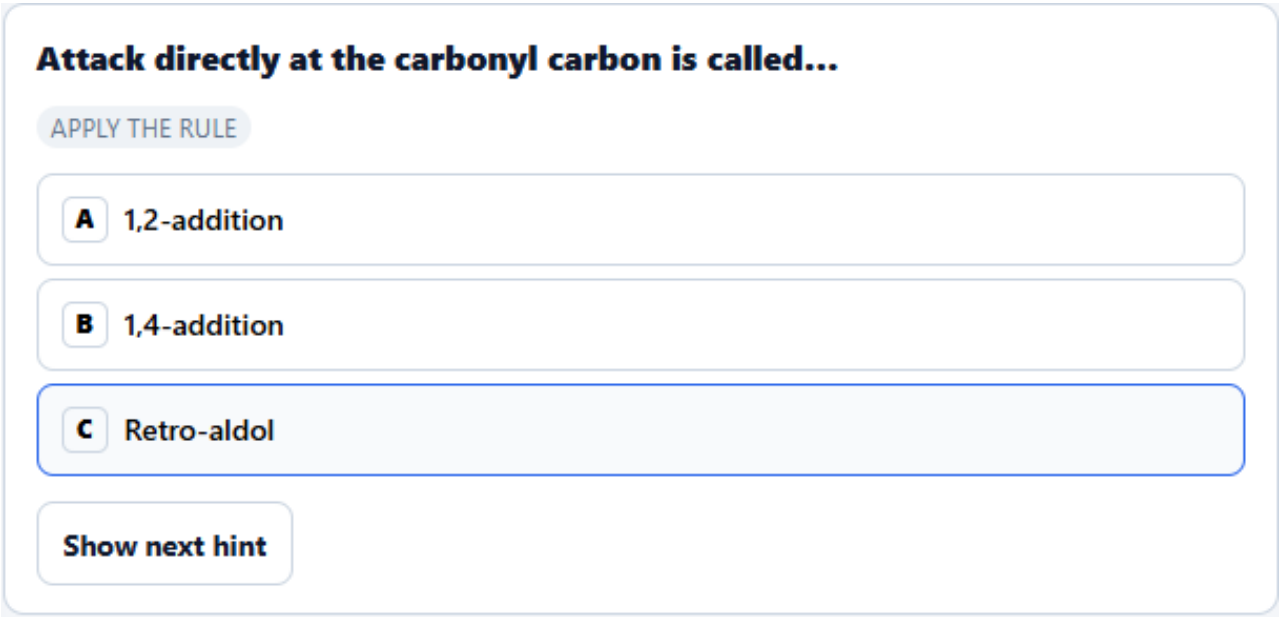
Option 1 (B): 1,4-addition

Option 2 (C): Retro-aldol

Screenshot: michael-site-stage-3-molecule



Screenshot: michael-site-stage-3-question



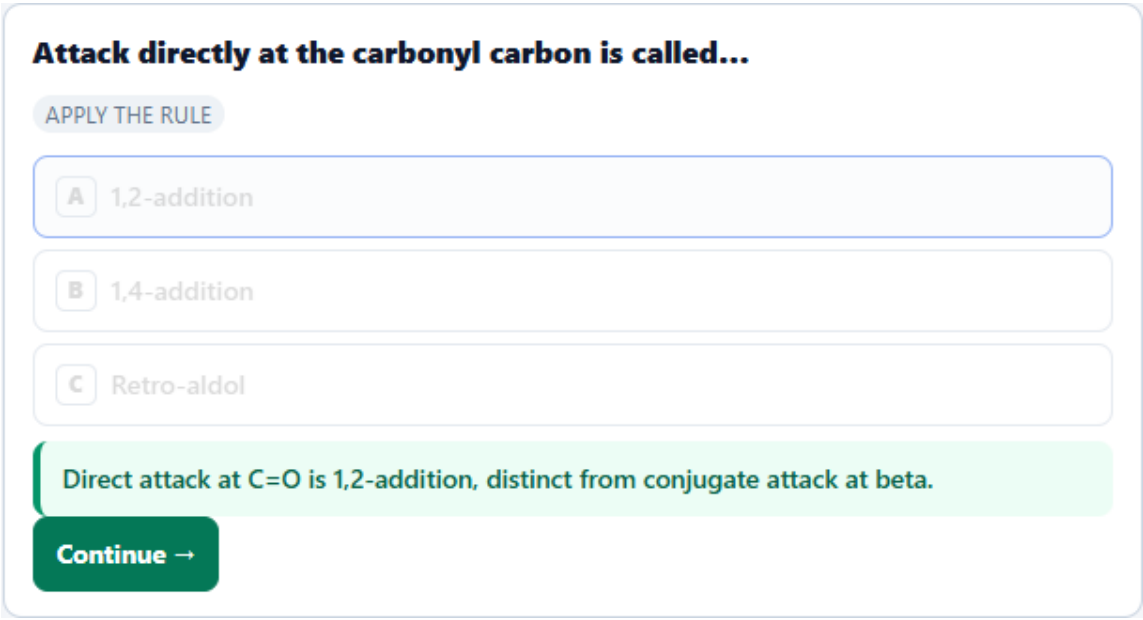
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Track the attacked site.

Hint 2: The oxygen and carbonyl carbon are positions 1 and 2 in this shorthand.

Explanation: Direct attack at C=O is 1,2-addition, distinct from conjugate attack at beta.

Screenshot: michael-site-stage-3-feedback

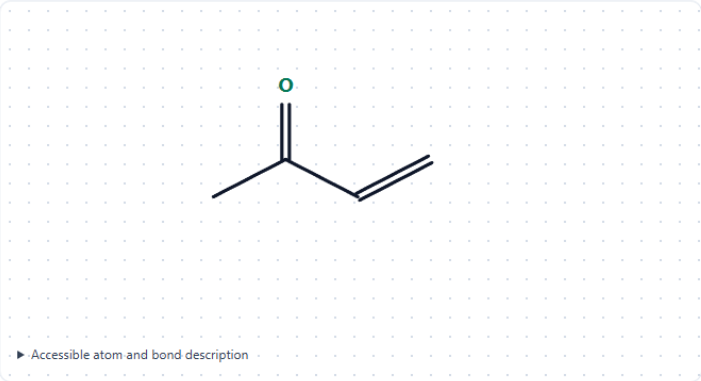


Completion summary: Conjugate addition attacks beta carbon.

Screenshot: michael-site-complete

Find a conjugate-addition site

Solved independently



▶ Accessible atom and bond description

`Conjugate addition attacks beta carbon.`

Formula: C4H6O

Replay workshop | Next workshop →

1. Select the carbonyl carbon.

B

> B is the carbonyl carbon.

2. Select the beta carbon attacked in a Michael addition.

D

> D is the terminal beta carbon.

3. Attack directly at the carbonyl carbon is called...

1,2-addition

> Direct attack at C=O is 1,2-addition, distinct from conjugate attack at beta.

Molecular repository: author claims to verify

ketone

Author claims: pentan-2-one | formula C5H10O

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=210.00, y=85.00; charge=0; explicit H override=none

4 (E): C; x=264.00, y=115.00; charge=0; explicit H override=none

5 (F): O; x=102.00, y=30.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [1, 5, 2]]

ethanal-enolate

Author claims: ethanal enolate (oxygen-charged contributor) | formula C2H3O

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=120.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=175.00, y=120.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1]]

vinylalcohol

Author claims: ethenol (vinyl alcohol) | formula C2H4O

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=165.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1]]

aldol

Author claims: 3-hydroxybutanal (stereochemistry unspecified) | formula C4H8O2

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

4 (E): O; x=50.00, y=45.00; charge=0; explicit H override=none

5 (F): O; x=166.00, y=175.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [0, 4, 2], [2, 5, 1]]

crotonaldehyde

Author claims: but-2-enal (E/Z unspecified) | formula C4H6O

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

4 (E): O; x=50.00, y=45.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [2, 3, 1], [0, 4, 2]]

methylvinylketone

Author claims: but-3-en-2-one | formula C4H6O

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

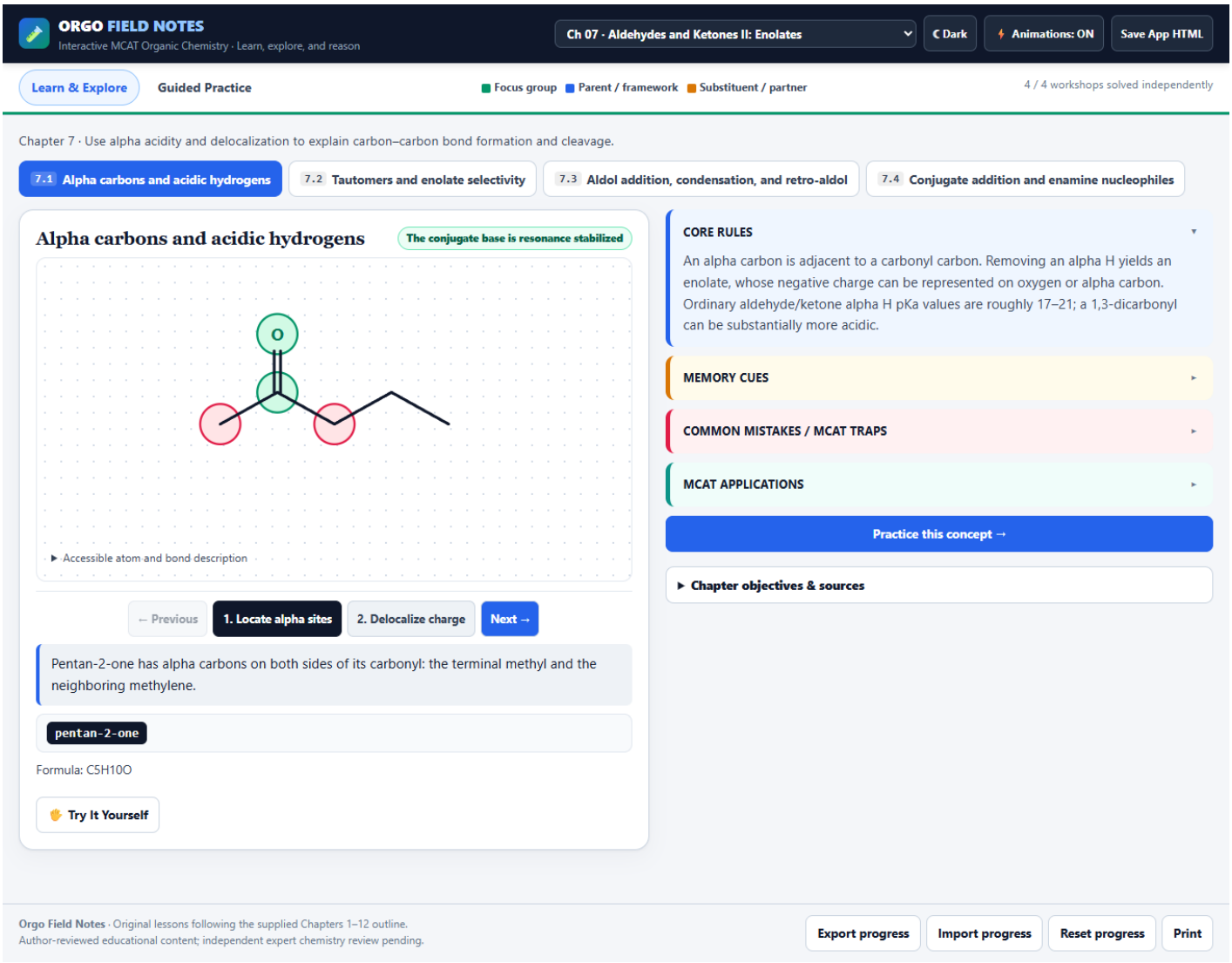
4 (E): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [1, 4, 2]]

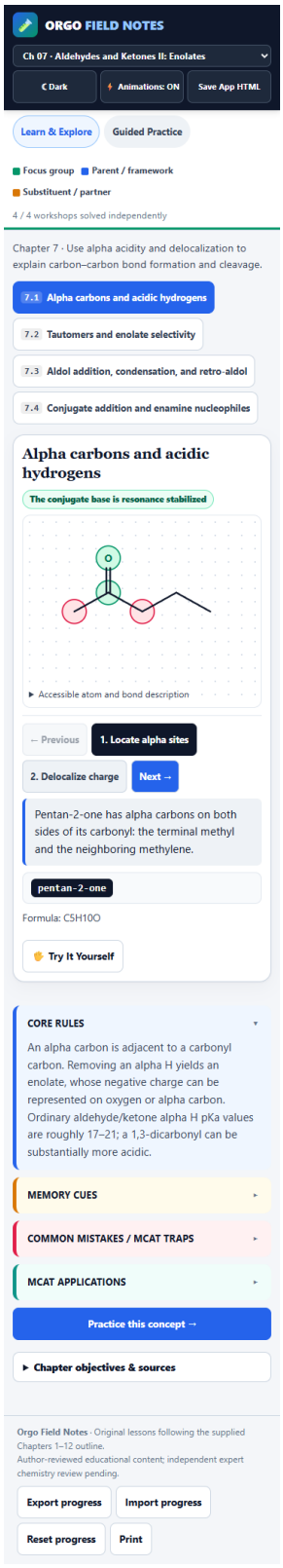
Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

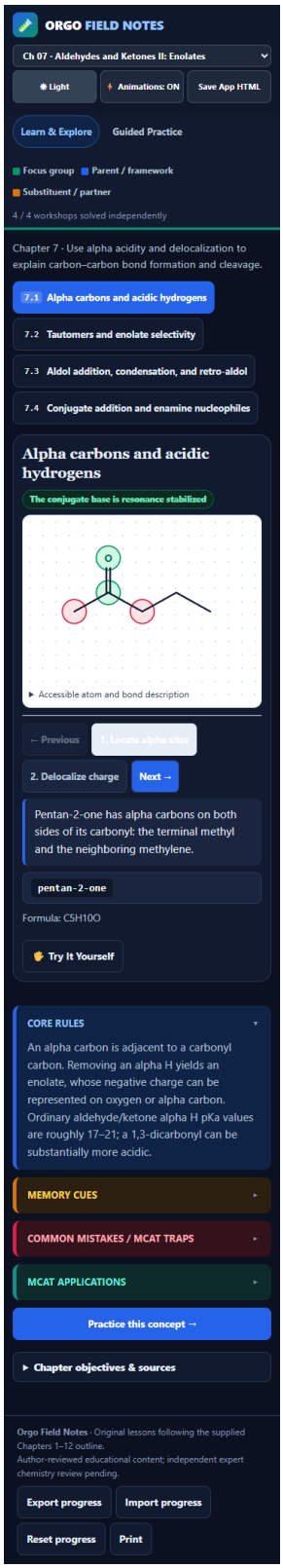
Screenshot: layout-desktop-light



Screenshot: layout-mobile-light



Screenshot: layout-mobile-dark



Execution evidence

55 screenshots and 14 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.